

NV-Reason-CT: 3D Visual Language Model for CT Analysis

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We present NV-Reason-CT, a generative vision–language model for chest and abdominal CT that combines native 3D visual encoding with radiologist-guided reasoning. CT interpretation requires integrating spatially distributed findings into a coherent account of anatomy, disease, and diagnostic uncertainty. Our model couples a native 3D vision transformer with a language model, passing all encoded visual tokens and their explicit 3D coordinates into language decoding without further spatial token merging. This design retains volumetric spatial information both within the vision encoder and through the language model’s positional encoding during joint processing with text.

We train on a curated corpus of approximately 550,000 multimodal instruction examples derived from 70,111 unique CT image inputs. The corpus combines standardized reports, abnormality-focused and anatomy-specific questions, multi-turn interactions, and radiologist-authored reasoning collected through recorded and transcribed expert CT interpretations. These expert annotations provide direct supervision and guide the generation of additional report-grounded synthetic reasoning. End-to-end supervised fine-tuning (SFT) is followed by Group Relative Policy Optimization (GRPO), using verifiable rewards over chest and abdominal abnormality sets.

The model supports direct abnormality classification, report generation, and interactive reasoning, presenting reviewable observations, differential diagnoses, and uncertainty alongside its conclusions. Evaluation spans public CT benchmarks and a held-out NIH cohort. On CT-RATE, NV-Reason-CT achieves a macro-F1 of 0.614 and a macro-AUROC of 0.871 without a task-specific classification head; generated reports achieve a report-derived macro-F1 of 0.592. In a preliminary study with expert radiologists, AI-assisted review received favorable confidence ratings and was associated with a 50% reduction in average reported interpretation and reporting time. We release the model and training code to support reproducible research on explainable AI for volumetric medical imaging.

GitHub Code: <https://github.com/NVIDIA-Medtech/NV-Reason-CT>

Model: <https://huggingface.co/nvidia/NV-Reason-CT>

1. Introduction

Computed tomography (CT) is intrinsically volumetric. A single examination can contain hundreds of cross-sectional images, and clinically important evidence depends on the morphology, extent, laterality, and anatomical distribution of findings across slices. Interpretation requires connecting these observations across a 3D volume and synthesizing them into a coherent account of normal and abnormal anatomy, diagnostic uncertainty, and an actionable impression. For a vision–language model, this creates two connected requirements: representing volumetric anatomy and communicating the clinical significance of the observed findings. Figure 1 illustrates this volumetric image-to-language objective.

Recent CT foundation models have demonstrated the value of learning from volumetric images and associated clinical text. CT-CLIP and CT-CHAT [1], Merlin [2], and COLIPRI [3] provide important foundations for CT representation learning, abnormality recognition, and generative interpretation. Within these systems, the interface between the visual encoder and language model is an important architectural

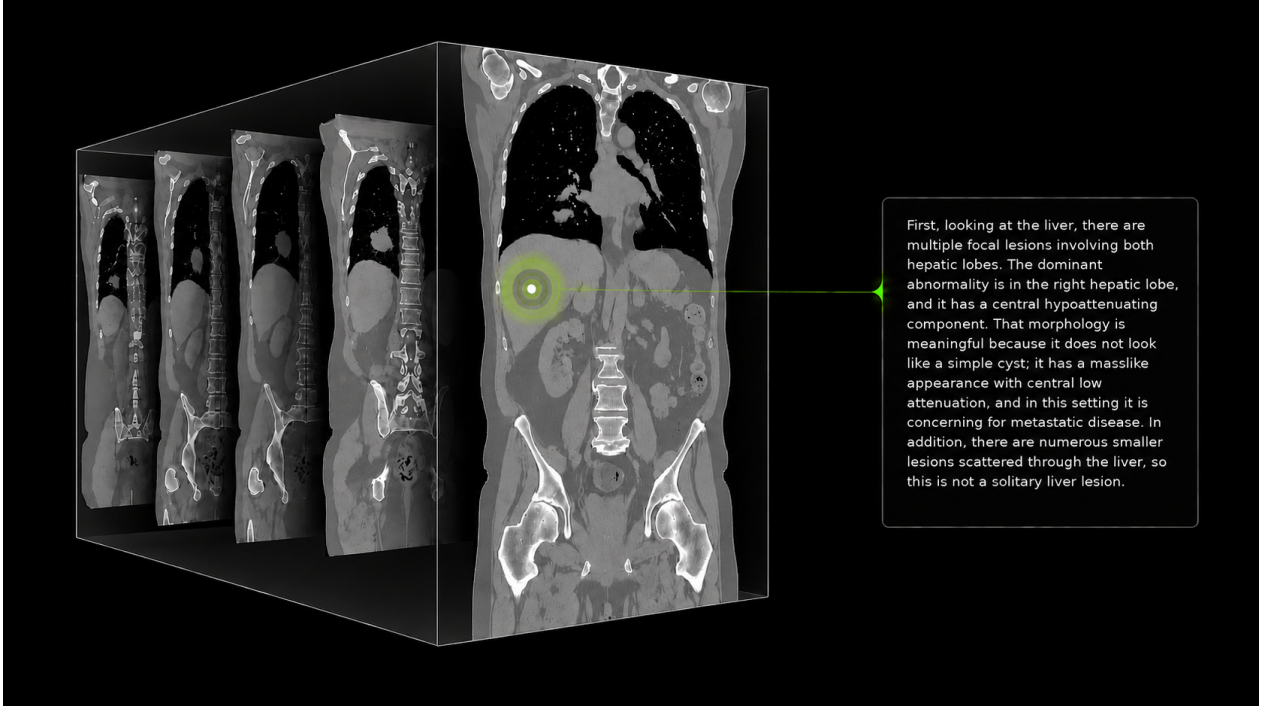


Figure 1 | A volumetric 3D CT examination is provided as a native 3D input, and the model produces an image-conditioned natural-language interpretation. The displayed target marker and example narrative are illustrative.

choice. Pooling or resampling can reduce the spatial granularity of the visual representation, while serializing features does not automatically retain their explicit 3D coordinates in the language decoder. Although visual features can encode geometry, preserving their depth, height, and width indices provides an additional means of representing spatial relationships during joint processing with text. This motivates an interface that retains both volumetric features and their spatial indexing.

Clinical assistance also requires communicating how observations support an interpretation. Radiologists organize their review around anatomical structures, relate findings to diagnostic possibilities, acknowledge uncertainty, and revisit the examination in response to follow-up questions. These activities motivate supervision that includes the observations and diagnostic considerations accompanying a report. We treat generated reasoning as a clinical explanation that can be reviewed alongside the image and final answer. Its value depends on factual grounding and usefulness to the reader; a plausible explanation alone does not establish that it faithfully represents the model’s internal computation [4].

Our previous work, NV-Reason-CXR-3B [5], introduced a reasoning-centered framework for chest radiography using radiologist-authored demonstrations, report-grounded synthetic reasoning, and reinforcement learning with verifiable abnormality rewards. The present work extends this methodology to chest and abdominal CT. This extension requires connecting expert-guided interpretation with a native volumetric architecture and accounting for differences in anatomical coverage, report structure, and finding vocabularies across regions.

Here we present NV-Reason-CT, a generative vision-language model that combines native 3D visual encoding with radiologist-guided supervision and verifiable reinforcement learning. The model couples the Qwen3.5-4B language model [6] with a native 3D vision transformer initialized from COLIPRI pretrained weights [3]. It processes regional chest or abdominal volumes and passes all encoded visual tokens through a learned projector to the language model without further spatial token merging. Explicit depth, height, and width coordinates are incorporated through the decoder’s positional encoding, retaining volumetric spatial indexing during joint processing of visual and text tokens.

We construct a curated training corpus of approximately 550,000 multimodal instruction examples from

70,111 unique CT image inputs. A limited set of recorded and transcribed expert-narrated CT interpretations provides direct supervision and examples of systematic anatomical review, diagnostic considerations, and uncertainty. These expert examples also guide the rewriting of source reports into synthetic narrated interpretations, with the source reports constraining the clinical content. The resulting corpus combines expert and synthetic narrations with standardized reports, finding- and anatomy-specific questions, multi-turn interactions, and refusal examples. Its construction involves extensive rewriting and curation of structured reports, synthetic narrations, and question–answer pairs, together with harmonization of finding terminology and anatomical coverage. This process broadens the available supervision while preserving the distinction between expert-authored examples and report-derived synthetic content.

Training proceeds in two stages. End-to-end supervised fine-tuning (SFT) jointly adapts the pretrained 3D encoder, learned projector, and language model to the curated tasks. Subsequent Group Relative Policy Optimization (GRPO) [7] uses region-specific abnormality and report-structure rewards to optimize finding agreement while maintaining the intended reporting format. This enables reinforcement learning from verifiable finding labels without requiring an expert-authored reasoning trace for every training volume.

We evaluate abnormality classification and report generation across CT-RATE, Merlin, RAD-ChestCT, and a held-out NIH cohort, complemented by qualitative analysis of reasoning and follow-up interaction and a preliminary assessment by expert radiologists. Together, these evaluations examine finding recognition, the clinical content of generated reports, and the usefulness of explanations for human review. We release NV-Reason-CT and its SFT and GRPO training code to support reproducible research on explainable AI for volumetric medical imaging.

2. Related work

Large-scale image–text learning and visual instruction tuning have made open-ended image interpretation increasingly feasible. Vision Transformers [8] represent images as patch tokens; CLIP [9] demonstrated that natural-language supervision can produce transferable visual representations; and models such as Flamingo [10] and LLaVA [11] showed how visual features can condition a general-purpose language model. Medical generalist models such as Med-PaLM M [12] have extended this paradigm to clinical question answering and report generation. VILA-M3 [13] further showed how knowledge from domain-specific medical expert models can be incorporated into a medical VLM. Most of this progress, however, has been developed around two-dimensional inputs. Applying the same interface to a CT by selecting a small number of slices, processing slices independently, or aggressively resampling visual features creates a bottleneck precisely where volumetric context is most important.

Early volumetric CT foundation models primarily pursued representation learning: they sought an image encoder whose features transferred across classification and other downstream tasks, rather than an end-to-end system for open-ended clinical generation. Most followed the CLIP paradigm by aligning CT volumes with text derived from radiology reports or diagnostic labels [1, 2, 3, 14]. A parallel line of work learned image representations without paired text, commonly through self-supervised objectives such as DINO [15]. More recent systems have increasingly added dedicated report decoders or connected CT encoders to general-purpose language models. We organize this progression into representation-learning methods, CT report-generation systems, and generative CT vision–language models. These categories overlap because a strong representation encoder can later serve as the visual component of a generative system.

2.1. Contrastive and representation-learning methods

This class of methods is defined primarily by how the visual representation is learned. Contrastive approaches jointly embed a CT volume and its associated report, report sentence, or label description, encouraging matched image–text pairs to be close and unmatched pairs to be separated. The resulting encoder can support zero-shot classification by comparing an image with positive and negative text prompts, or it can be transferred to a specific task by training a linear or task-specific prediction head on top of its features. Although the text encoder provides supervision during contrastive pretraining, it is not always part of the deployed model. For linear-probe or task-specific transfer, the text encoder is typically discarded and only the trained image encoder is retained. It remains necessary for zero-shot image–text scoring and cross-modal retrieval, where it encodes the comparison prompts or reports [1, 14, 16]. Image-only self-supervised methods pursue the

same goal of a reusable CT representation but replace paired text with objectives such as teacher–student consistency or masked-image learning. Either type of encoder can subsequently be coupled to a report decoder or an LLM, so the contrastive or self-supervised objective describes the representation-learning stage rather than the full capabilities of the final system.

CT-CLIP. CT-CLIP introduced large-scale contrastive alignment between volumetric chest CT and radiology reports using the CT-RATE training set [1]. This patient-disjoint training cohort contains 47,149 reconstructed chest CT volumes from 24,128 studies and 20,000 patients. Its factorized vision transformer first models in-plane patches and then aggregates through-plane information. The native classification benchmark covers all 18 CT-RATE abnormalities through paired positive and negative text prompts. CT-CLIP also provides the visual foundation for CT-CHAT, which follows a LLaVA-style design: attention pooling compresses the encoder output into 256 image tokens, and an MLP maps them into the Llama 3.1 embedding space for instruction following and report generation. The language model receives this 1D token sequence without an explicit depth–height–width coordinate grid.

Merlin. Merlin learns an abdominal CT representation by aligning a 3D ConvNet with reports and structured clinical data [2]. Representation learning uses 15,331 abdominal CTs from the Stanford cohort, together with their reports and approximately 1.8 million structured clinical codes. Its native zero-shot classification task contains 30 report-derived findings, with unmentioned labels excluded and positive and explicitly negative cases balanced separately for each finding. Although its main contribution is representation learning, Merlin also attaches a language model for section-wise abdominal report generation. A linear adapter projects the encoder’s last $7 \times 7 \times 10$ feature grid into RadLlama-7B input tokens; the resulting flattened sequence does not retain an explicit 3D coordinate grid inside the language model.

COLIPRI. COLIPRI combines CT–report contrastive learning with image-only self-supervision, opposite-sentence discrimination, and report-generation objectives in a native 3D vision transformer [3]. It is developed entirely on chest CT: paired vision–language training uses 22,676 CT-RATE volumes, while image-only self-supervision additionally uses approximately 72,000 low-dose NLST scans. Its CT-RATE classification experiments use the original 18-label ontology, while its RAD-ChestCT experiments report 16-label results without fully enumerating the mapping. COLIPRI-CRM further connects the visual encoder to a Qwen2.5 language model for report generation through a two-layer MLP following the LLaVA framework. Qwen2.5 receives the projected patch embeddings as a 1D token sequence rather than with their original depth–height–width coordinates. Our model uses the released COLIPRI Primus encoder weights as a strong initialization, rather than as a permanently frozen feature extractor; the vision encoder, multimodal merger, and language model are subsequently optimized together in our end-to-end training.

Pillar-0. Pillar-0 introduced Atlas, a hierarchical volumetric encoder whose multiscale attention makes processing full-resolution CT and MRI volumes computationally efficient [14]. Its four modality-specific encoders are pretrained on chest, abdomen–pelvis, and head CT examinations and breast MRI examinations from one academic-center archive. The encoders receive multiple window-level views of each scan and are pretrained with an asymmetric CLIP-style objective that aligns the Atlas volume representation with a frozen Qwen3-Embedding-8B text encoder. Pillar-0 is a non-generative representation model: downstream tasks use frozen embeddings with learned prediction heads.

Jolia. Jolia adopts Atlas as its principal visual backbone and follows Pillar-0’s use of multi-window CT inputs and a frozen Qwen3-Embedding-8B text encoder, but it is trained as a separate model [16]. Its headline configuration uses a smaller Atlas encoder trained from scratch on 74,434 public CT–report pairs from CT-RATE and INSPECT chest scans and Merlin abdominal scans. The study also evaluates 3D ResNet-101 and ViT-B backbones. Jolia’s main innovation, ConQuer, augments global CLIP alignment with learnable anatomy-specific queries and a separate contrastive loss for each concept, producing localized visual representations without segmentation supervision. It supports 18-label CT-RATE classification and, like Merlin, also adds a language-model decoder for report generation. That pathway projects a single pooled 576-dimensional global image feature into one Qwen3.5 input token; the localized ConQuer tokens and their 3D relationships are not passed to the language model.

VoxelFM. VoxelFM is an image-only 3D transformer pretrained with DINO-style self-supervision and adapted with lightweight supervised heads [15]. Pretraining uses 137,107 CT studies spanning the thorax, abdomen, and head and neck, drawn from CT-RATE, INSPECT, Merlin, NLST, and other public collections. Its downstream classifiers cover all 18 CT-RATE findings and all 30 Merlin findings, and a separate Q-Former

plus Qwen3-8B pathway supports CT-RATE report generation. The Q-Former converts the patch features into learned query outputs that the language model consumes as a 1D sequence, without an explicit dense 3D coordinate grid.

CT-SSG. CT-SSG uses a supervised 2.5D convolutional encoder with graph-based slice aggregation [17]. It is trained and evaluated by fivefold cross-validation on the 25,692-acquisition CT-RATE chest cohort, retaining one reconstruction per acquisition.

fVLM. fVLM learns organ-conditioned contrastive representations using segmentation-derived anatomical masks [18]. Its pretraining resource, MedVL-CT69K, contains 272,124 multiregion CT scans from 69,086 patients, with 64,476 patients assigned to training; it includes chest, abdomen, pelvis, and other body regions.

2.2. CT report-generation systems

Dedicated CT reporting systems differ in how directly report text is generated from the image and in how clinical content is measured.

CT2Rep. CT2Rep generates CT-RATE findings with a volumetric encoder and relational-memory transformer decoder [19]. It is trained on the CT-RATE chest training split, comprising 47,149 reconstructed volumes from 20,000 patients.

CT-CHAT. CT-CHAT connects the CT-CLIP encoder to Llama 3.1 through a learned multimodal projector [1, 20]. Its projector and language-model adapters are trained on more than 2.7 million question-answer pairs derived from the CT-RATE chest training cohort.

BTB3D. BTB3D first compresses the volume into discrete causal 3D tokens before language decoding [21]. Its report generator is likewise trained on the 47,149 reconstructed chest volumes in the CT-RATE training split.

These systems are evaluated on CT-RATE through the 18 report-derived abnormalities; CT-CHAT and BTB3D are also evaluated on the 16-label RAD-ChestCT mapping after abnormalities are extracted from generated reports.

MS-VLM. MS-VLM combines a self-supervised 2D slice encoder with an inter-slice transformer and a language-model bridge [22]. Its chest model uses 47,149 CT-RATE training volumes, approximately 180,000 synthetic QA pairs, and selected QA from the 1.3-million-pair RadGenome-ChestCT resource; a separate rectal MRI experiment uses 186 training patients.

CT-AGRG. CT-AGRG takes a more explicitly label-driven approach: an 18-label CT-RATE classifier first selects positive abnormalities, after which a decoder generates one sentence for each selected finding [23]. It is trained on an earlier CT-RATE chest split containing 34,781 reconstructed volumes from 17,799 patients.

These designs illustrate an important distinction between open-ended image-conditioned reporting and reports whose content is gated by a separately supervised abnormality classifier.

DCP-PD. DCP-PD uses the Pillar-0 Atlas encoder and can condition its report generator on CT-RATE-trained discriminative predictions [24]. It uses the official 24,128-study CT-RATE chest training partition after removing approximately 800 head-only studies.

EXACT-CHAT. EXACT-CHAT derives 18 voxel-level anomaly maps and passes their diagnostic predictions to its report decoder; its RAD-ChestCT evaluation uses a modified 16-label mapping [25]. Its weakly supervised vision pretraining and report-generation training use 24,128 CT-RATE chest volumes.

MonteRET. MonteRET retrieves condition- and region-matched cases to support report generation and refinement [26]. It is trained on 24,128 RadGenome-ChestCT scans—the CT-RATE chest training studies paired with anatomically organized reports—and uses the same training reports as its retrieval database.

Astra. Astra combines a Merlin encoder with supervised and reinforcement-learning stages [27]. It is developed on CTRgDB, a harmonized corpus of 90,678 thoracoabdominal CT-report pairs assembled from CT-RATE, INSPECT, Merlin, AbdomenAtlas 3.0, and BIMCV.

Resolution Meets Reduction. Resolution Meets Reduction couples a high-resolution slice encoder to a language model through a Perceiver resampler [28]. Its controlled report-generation experiments train

separately on 22,590 CT-RATE chest studies and 19,361 Merlin abdominal cases.

The CT-RATE clinical-content results reported for these systems are generally derived from the original 18-label ontology, although evaluation units, label extractors, and averaging conventions differ.

As noted above, Merlin and Jolia straddle the representation and reporting categories. We retain them in the first group because their principal novelty is the learned CT representation, while explicitly recognizing that both also attach language-model decoders through the projected, non-grid-aware interfaces described above and provide report-generation results.

2.3. Generative CT vision–language models

General-purpose medical vision–language models aim to support classification, question answering, and free-form generation with a common interface.

RadFM. RadFM uses a 3D vision transformer and Perceiver-style resampler with an autoregressive medical language model [29]. It is pretrained on MedMD, a broad multimodal mixture containing 15.5 million 2D images and 500,000 3D scans across CT, MRI, PET, radiography, and anatomical regions from brain to pelvis, then fine-tuned on a three-million-pair radiology subset.

M3D-LaMed. M3D-LaMed combines a native 3D transformer with staged image–text alignment and multimodal instruction tuning [30]. Training uses M3D-Data, which contains 120,092 CT image–text pairs covering anatomy throughout the body and collected from Radiopaedia, plus 662,000 instruction–response pairs spanning report generation, VQA, localization, and segmentation.

Both models can be prompted independently for all 18 CT-RATE abnormalities; later studies also evaluate their generated reports against 16-label RAD-ChestCT targets.

MedGemma 1.5. MedGemma 1.5 represents a CT as a sequence of sampled 2D slices and uses a general medical multimodal decoder [31]. Its added volumetric training data include 282,963 internal CT examples covering head, chest, and abdomen, alongside 167,674 multiregion MRI examples and the broader MedGemma medical mixture.

ClinFusion. ClinFusion combines several 2D encoders with a native 3D pathway before fusing the representations into Qwen3-VL [32]. Its progressive training corpus contains approximately 22.2 million mixed samples across 12 medical imaging modalities; the volumetric stages draw on CT-RATE, Merlin, RadGenome-ChestCT, INSPECT, M3D-Cap, AMOS, and synthetic chest and abdominal CT collections.

We evaluate both model families on all 18 CT-RATE findings using per-abnormality binary questions. Their broader generative interfaces make them useful comparisons for instruction-following behavior, but their visual interfaces and spatial assumptions differ from native dense 3D token processing.

Chain-of-thought prompting [33] and reasoning-oriented reinforcement learning, exemplified by DeepSeek-R1 [34], have shown that language models can produce structured, multi-step solutions. In a medical setting, however, a generated rationale should not be assumed to reveal the model’s faithful internal computation [4].

A common design across language-connected CT systems is to project or resample visual features into a 1D sequence that the language model then consumes alongside text [1, 29, 30, 15, 28]. Some of these models encode 3D geometry inside the vision tower; this should not be confused with carrying the original 3D token coordinates through the language model itself. After the connector, the language decoder commonly receives flattened visual embeddings without an explicit dense depth–height–width grid in its own positional mechanism.

NV-Reason-CT. Our model is designed to preserve this structure at both stages. Its 3D vision transformer uses three-axis rotary positional encoding and produces a $24 \times 24 \times 24$ grid of 13,824 visual tokens. We initialize the Primus visual encoder [35] from COLIPRI pretrained weights [3] and connect it to Qwen3.5-4B [6] through a learned projector that retains every visual token without further spatial merging. The original grid dimensions accompany these tokens, allowing the language model’s multimodal rotary positional encoding (MRoPE) [36] to assign each token explicit depth, height, and width indices. Volumetric position therefore remains represented throughout language-decoder self-attention during joint processing with text, in addition to its representation within the visual encoder. This design is intended to support spatially specific interpretation,

such as distinguishing left- from right-sided renal lesions, describing craniocaudal extent, or localizing findings to different thoracic regions. End-to-end SFT jointly adapts the encoder, projector, and language model using curated chest and abdominal CT tasks, including radiologist-guided reasoning. Subsequent GRPO uses region-specific verifiable rewards to optimize abnormality agreement and report structure.

3. Methods

3.1. Overview

NV-Reason-CT combines a native 3D vision–language architecture with radiologist-guided supervision and verifiable reinforcement learning for chest and abdominal CT interpretation. The model represents a regional CT volume as spatially indexed visual tokens and processes them alongside a natural-language instruction to generate structured reports, finding-specific answers, and clinical explanations.

Training adapts the reasoning-centered strategy of NV-Reason-CXR [5] to volumetric CT. A limited set of recorded and transcribed expert interpretations provides direct supervision and guides the construction of synthetic narrated interpretations from source reports. These examples form part of a curated corpus that also includes standardized reports, abnormality questions, localization and attribute questions, follow-up interactions, and refusal examples. End-to-end supervised fine-tuning (SFT) jointly adapts the visual encoder, projector, and language model to these complementary tasks. Subsequent Group Relative Policy Optimization (GRPO) [7] uses region-specific abnormality labels and output-structure requirements to optimize finding agreement and report organization without requiring an expert-authored reasoning trace for every training volume.

Generated reasoning is presented as a reviewable clinical explanation that organizes observations, relevant normal findings, uncertainty, and conclusions. Its usefulness must be assessed through factual grounding, report quality, downstream task performance, and expert evaluation; it is not assumed to reveal the model’s internal computation.

3.2. Model architecture and volumetric interface

NV-Reason-CT couples a Primus 3D vision transformer [35], initialized from released COLIPRI pretrained weights [3], with the Qwen3.5-4B language model [6] through a learned projector. Figure 2 summarizes the architecture and its volumetric interface.

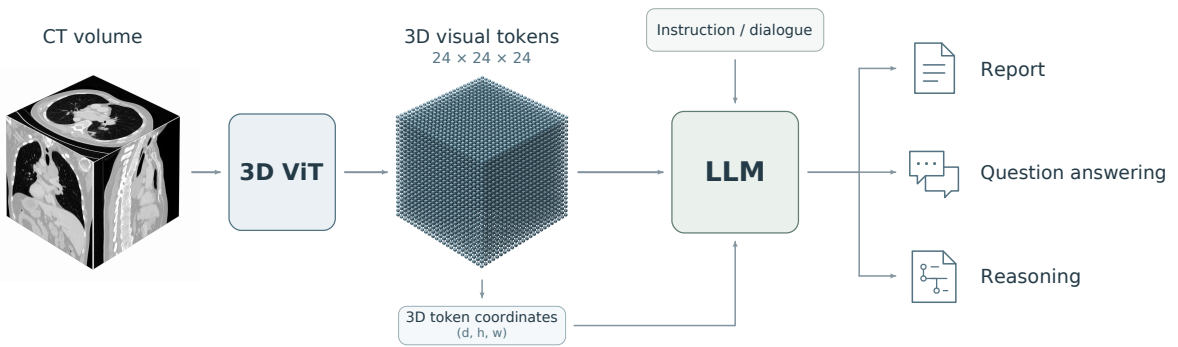


Figure 2 | Architecture overview of NV-Reason-CT. A native 3D visual encoder, shown together with its learned projection as the 3D ViT block, maps a CT volume to a $24 \times 24 \times 24$ grid of visual tokens. The LLM receives the projected tokens and instruction or dialogue context. Each visual token is assigned depth, height, and width coordinates (d, h, w) derived from the token grid and incorporated through multimodal rotary positional encoding (MRoPE). Outputs include reports, question answering, and reasoning explanations. The CT cube is an illustrative rendering based on orthogonal views extracted from a CT scan.

For a preprocessed single-channel volume $x \in \mathbb{R}^{1 \times 192 \times 192 \times 192}$, the visual encoder uses non-overlapping

$8 \times 8 \times 8$ -voxel patches and three-axis rotary positional encoding to produce a $24 \times 24 \times 24$ grid of 864-dimensional features. The projector maps each feature to the language model’s 2,560-dimensional embedding space without spatial merging, preserving all 13,824 visual tokens.

The projected tokens are serialized for language-model input, accompanied by their original grid dimensions. The language model’s multimodal rotary positional encoding (MRoPE) [36] uses these dimensions to assign each visual token explicit depth, height, and width indices. Volumetric position is therefore represented both within the visual encoder and during language-decoder self-attention as visual and text tokens are processed together. This interface retains the spatial indexing of the encoded volume through language decoding.

3.3. Volumetric preprocessing

Each NIfTI volume is loaded as a single-channel tensor and processed using MONAI transforms [37]. Volumes are reoriented to left–posterior–superior (LPS) orientation and resampled to 2-mm isotropic spacing using trilinear interpolation. After resampling, dimensions smaller than 192 voxels are padded with $-1,000$ Hounsfield units (HU) before the anatomy-aware region selection described in the following subsection. The resulting $192 \times 192 \times 192$ -voxel input corresponds to a $384 \times 384 \times 384$ -mm field of view. Intensities are clipped to $[-1000, 1000]$ HU and divided by 1,000, yielding values in $[-1, 1]$.

3.4. Anatomy-aware cropping: chest and abdomen

An anatomical region of interest could be obtained using an organ-segmentation model or a learned 3D bounding-box detector. Both approaches introduce an additional model into preprocessing, with associated integration, computational, and validation requirements. Here, we use an automatic, deterministic heuristic to select regional inputs for the chest and abdomen, the two anatomical regions represented during training. Each input is assigned a chest or abdominal region designation that determines the cropping procedure.

After resampling, the method thresholds the CT at -500 HU to form a body mask. Slice-wise morphological hole filling identifies low-attenuation air enclosed by the body, and 3D connected-component analysis groups these regions. Component selection uses size and superior location to identify candidate lung regions, which are accepted only when their combined physical volume is at least 250 mL.

The detected lung bounds provide a common anatomical landmark for both supported regions. For chest inputs, the $192 \times 192 \times 192$ -voxel crop is centered craniocaudally on the detected lungs. For abdominal inputs, a target interval is defined using a 300-mm length extending primarily inferiorly from the lung base, with up to 100 mm of overlap above it, subject to the available anatomical coverage. This interval determines the center of the fixed $192 \times 192 \times 192$ -voxel crop. The in-plane crop is centered in both cases.

When lung bounds cannot be recovered, the preprocessor uses a deterministic edge-aligned fallback crop. The heuristic depends on sufficient lung visibility and may mislocalize the intended region when lung landmarks are absent or difficult to identify. Figure 3 illustrates chest and abdominal crop placement in the same CT volume. Additional views of these crops and their corresponding model outputs are shown in Appendix D.

3.5. Problem formulation and output representation

Let x denote a regional CT volume, q the instruction and any preceding dialogue context, and y the model response. A single conditional policy $\pi_\theta(y \mid x, q)$ supports structured chest and abdominal reports, free-form or constrained question answering, and multi-turn follow-up interaction. For structured-report reinforcement learning, each example also includes an anatomy region $a \in \{\text{chest}, \text{abdomen}\}$ and a reference finding set $z \subseteq \mathcal{A}_a$, where $\mathcal{A}_a$ is the corresponding 30-finding chest or 29-finding abdominal ontology described in Section 4.2.

Reasoning-enabled SFT responses place the clinical analysis within a `<think>...</think>` span. In the structured-report format used for GRPO, the report is followed by exactly one machine-readable `<answer>...</answer>` block containing a comma-separated finding set. This block provides the deterministic finding representation used to compute the abnormality-agreement reward. Other SFT tasks retain their task-specific response formats, including free-text and Yes/No answers.

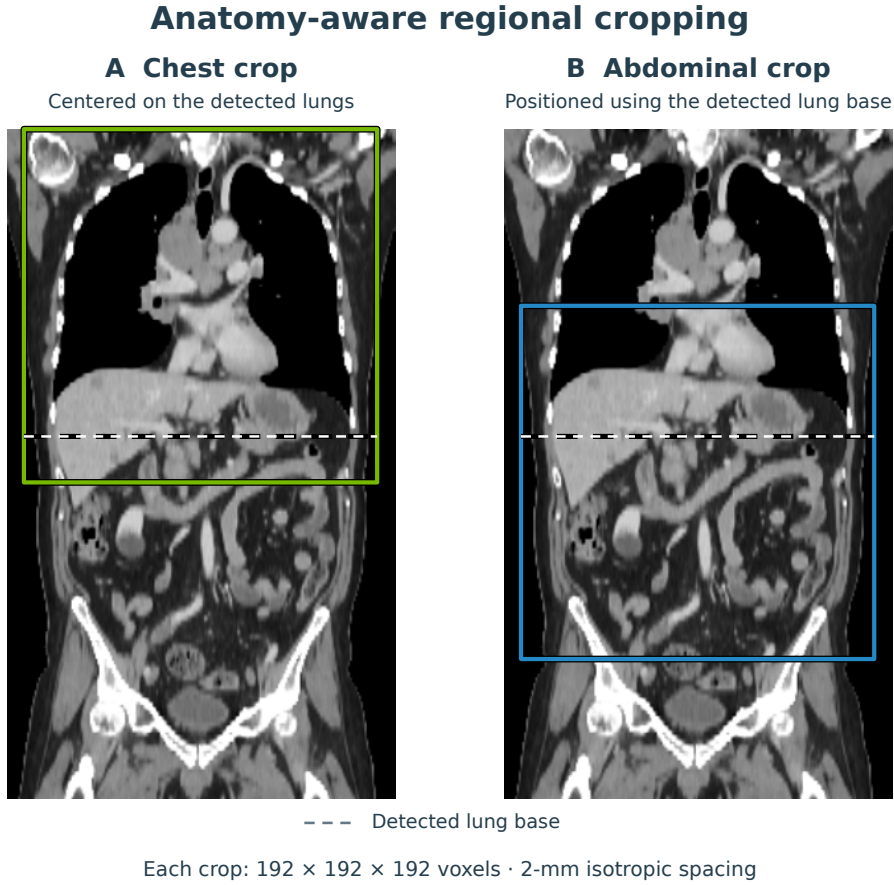


Figure 3 | Anatomy-aware regional cropping. The same coronal CT slice is shown at the same scale and display window with the chest crop outlined in green (A) and the abdominal crop in blue (B). Dashed lines mark the inferior bound of the detected 3D lung region, which provides a landmark for crop placement. Each square outlines a cross-section through a $192 \times 192 \times 192$ -voxel cubic input at 2-mm isotropic spacing (384 mm per side). The model processes one regional crop at a time.

3.6. Supervised fine-tuning

We jointly fine-tune the 3D visual encoder, projector, and language model on the curated instruction corpus. Figure 4 summarizes the training workflow from radiologist-guided data curation through SFT and GRPO.

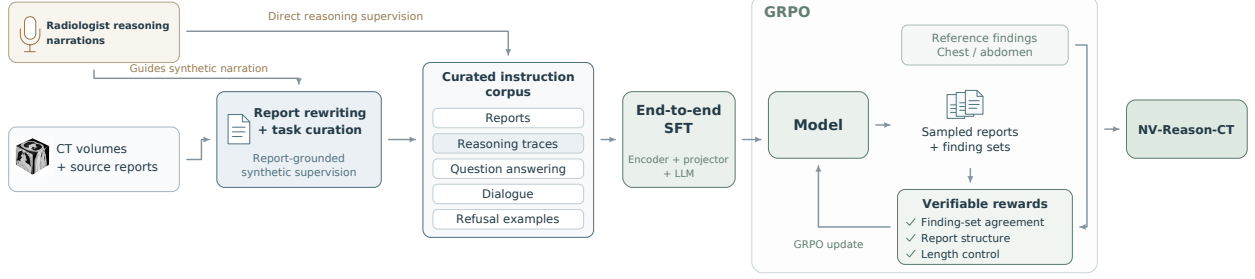


Figure 4 | Training workflow for NV-Reason-CT. Radiologist reasoning narrations provide direct supervision and guide source-report rewriting into synthetic narrations. Together with curated reports, question answering, dialogue, and refusal examples, they form the instruction corpus for end-to-end SFT of the visual encoder, projector, and language model. Starting from the SFT model, GRPO samples reports and finding sets for CT-conditioned prompts and scores each completion using region-specific reference findings, report structure, and completion length. The resulting rewards guide policy updates.

Given an SFT example (x, q, y^*) , where y^* is the target assistant response, we minimize the autoregressive cross-entropy:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x, q, y^*)} \sum_{t=1}^T \log \pi_{\theta}(y_t^* | x, q, y_{<t}^*).$$

Here, T is the target-response length. System and user tokens are excluded from the loss. In multi-turn examples, preceding dialogue provides conditioning context, and supervision is applied only to the final assistant response.

The training mixture includes region-specific and generic structured-report prompts, expert and report-derived synthetic reasoning examples, concise report-planning rationales, abnormality lists, binary and section-specific questions, location, laterality, and severity questions, single- and multi-turn follow-ups, and refusal examples. Data sources, regional composition, and task-specific curation are described in Section 4.

3.7. Group Relative Policy Optimization

Starting from the SFT policy, we apply GRPO [7] to optimize structured report generation using region-specific abnormality and output-structure rewards. GRPO uses the same training split paired with reference abnormality sets. The model is prompted to generate a complete report followed by a finding-set answer block, and each completion is scored using the reward function defined in Section 3.8.

For each prompt, we sample G completions $y_{1:G} \sim \pi_{\theta_{\text{old}}}(\cdot | x, q)$ and assign each completion a scalar reward $r_i = R(y_i, z, a)$. Here, $\pi_{\theta_{\text{old}}}$ denotes the policy used to generate the completions. The group-relative advantage is

$$A_i = \frac{r_i - \frac{1}{G} \sum_{j=1}^G r_j}{\sigma_r + \epsilon}.$$

The numerator subtracts the mean reward within the prompt group, while σ_r is the standard deviation of rewards across the distributed batch. The small positive constant ϵ stabilizes the division.

The policy is updated with the clipped GRPO objective

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\mathbb{E}_{i,t} [\min(\rho_{i,t} A_i, \text{clip}(\rho_{i,t}, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}) A_i)],$$

where $\rho_{i,t} = \pi_{\theta}(y_{i,t} | x, q, y_{i,<t}) / \pi_{\theta_{\text{old}}}(y_{i,t} | x, q, y_{i,<t})$ is the token-probability ratio between the policy being optimized and the sampling policy. The indices i and t denote completions and token positions, respectively.

We use $\epsilon_{\text{low}} = 0.20$ and $\epsilon_{\text{high}} = 0.28$. The reported run uses no explicit reference model Kullback–Leibler (KL) penalty ($\beta = 0$).

The resulting GRPO checkpoint is used in the reported evaluations.

3.8. Region-aware verifiable rewards

The anatomy region a determines the finding ontology and report schema used to score each completion. The reward combines abnormality-set correctness, structured-report conformity, and a soft completion-length penalty:

$$R(y, z, a) = 2.0 r_{\text{set}} + 0.5 r_{\text{struct}} + r_{\text{len}}.$$

Abnormality-set correctness. Let $\hat{z} = \Phi(y)$ be the comma-separated finding set parsed from the first `<answer>...</answer>` block. Agreement with the reference finding set z is measured using set F1:

$$r_{\text{set}}(y, z) = \frac{2|\hat{z} \cap z|}{|\hat{z}| + |z|}.$$

The correctness term is set to zero unless the completion contains exactly one opening and one closing answer tag forming a valid answer block. A present but empty block is mapped to the region-specific no-finding label, whereas a missing block is invalid. A prediction combining the no-finding label with one or more positive findings is treated as contradictory and also receives zero correctness reward.

Structured-report reward. The format reward $r_{\text{struct}} \in [0, 1]$ evaluates conformity to the report schema for the selected anatomy region. It combines common report-block presence and ordering (0.15), region-specific organ-heading presence and ordering (0.55), non-empty section and impression text (0.20), and a numbered or bulleted impression (0.10). The Technique section is optional. This term evaluates report organization and does not itself assess the clinical correctness of the prose.

Length-band penalty. A bounded length penalty discourages completions containing only the finding list and excessively long reports. The penalty is zero within the target length band and decreases linearly outside it, with its magnitude capped at one:

$$r_{\text{len}}(y) = \begin{cases} \max\left(-1, \frac{L(y)-180}{180}\right), & L(y) < 180, \\ 0, & 180 \leq L(y) \leq 900, \\ \max\left(-1, \frac{900-L(y)}{300}\right), & L(y) > 900, \end{cases}$$

where $L(y)$ is the completion length in tokens.

3.9. Training and implementation details

Training uses 16 nodes, each equipped with eight NVIDIA H100 GPUs. We use fused AdamW with a per-device batch size of one and two gradient-accumulation steps. The training stack is implemented in PyTorch using Hugging Face Transformers, TRL, Accelerate, and MONAI. Our training code is publicly available¹.

SFT uses a base learning rate of 2×10^{-5} , a cosine schedule with 3% warm-up, and gradient-norm clipping at 0.3. Component-specific learning rates are $0.1 \times$ the base rate for the 3D visual encoder, $5 \times$ for the projector, and $1 \times$ for the language model.

For GRPO, we sample 16 completions per prompt at temperature 1.0, with a maximum completion length of 1,300 tokens. We use batch-level reward scaling, a learning rate of 1×10^{-6} , five warm-up steps, a cosine schedule decaying to 5% of the initial learning rate, and gradient-norm clipping at 1.0.

¹<https://github.com/NVIDIA-Medtech/NV-Reason-CT>

4. Data

4.1. Datasets

Our training data comprise three sources—CT-RATE, CancerVerse, and an internal NIH dataset—each containing 3D CT volumes paired with radiology reports (see Table 1). Where the field of view permits, a volume contributes separate chest and abdominal crops for region-specific tasks. Crop overlap is generally greater in CT-RATE than in the more extensive NIH scans. For evaluation, we use CT-RATE for chest, separate NIH internal chest and abdomen cohorts, Merlin as an external abdominal dataset, and RAD-ChestCT as an external chest dataset.

Table 1 | NV-REASON-CT data cohorts. Chest and abdomen columns report unique regional crops used for each task and are not mutually exclusive.

Dataset	Training			Evaluation		
	Overall	Chest	Abdomen	Overall	Chest	Abdomen
CT-RATE	47,149 cases; 24,128 studies; 20,000 patients	46,203	20,243	3,039 public validation cases; 3,002 after correction; 1,564 studies; 1,304 patients	3,002	–
NIH internal	15,991 cases; 15,991 patients	12,742	12,243	5,092 cases; 5,092 patients	3,981	4,205
CancerVerse	22,720 cases; 13,778 patients	4,668	4,438	–	–	–

CT-RATE. CT-RATE is a public dataset for chest CT representation learning, report generation, and multi-label abnormality classification [1, 38]. The release contains 25,692 non-contrast CT studies from 21,304 patients, paired with radiology reports and report-derived labels for 18 abnormalities. Because an acquisition can include multiple reconstructed series, the 25,692 studies correspond to 50,188 three-dimensional cases. The patient-disjoint split contains 47,149 training cases from 20,000 patients and 3,039 public validation cases from 1,304 patients. We use the corrected validation manifest, which removes 37 non-chest volumes and retains 3,002 cases while preserving all 1,564 validation studies. For training, we generated 46,203 chest crops. We additionally selected 20,243 cases with a substantial upper-abdominal field of view and included the upper abdomen as a separate regional crop.

NIH internal collection. The NIH internal data contains 15,991 training cases, each from a unique patient. From this cohort, we generated 12,742 chest crops and 12,243 abdominal crops. Most CTs included almost the whole body field of view, allowing for proper chest and abdominal crops with only a small overlap. A held out set of 5,092 cases was used for validation (each from a unique patient). Of these, 3,981 contain the full chest and 4,205 contain the full abdomen. Reports in both deliveries were mapped to 30 chest and 29 abdominal findings.

CancerVerse. CancerVerse is a public dataset centered on 13 abdominal and pelvic malignancy types and pairs CT volumes with radiology reports and clinical metadata [39]. We use 22,720 CT–report pairs to generate 4,668 chest crops and 4,438 abdominal crops, each targeting the full corresponding anatomical region. The available reports contain cancer-focused sections rather than complete radiology reports. Because most anatomical structures are not mentioned, these reports are not suitable as targets for structured report generation. We therefore use CancerVerse only for targeted VQA on the available cancer-related abnormalities.

Merlin (external). The Merlin Abdominal CT Dataset was used only for evaluation; it contains 25,494 abdominal or abdominopelvic CT–report pairs from 18,317 patients treated at Stanford Hospital between 2012 and 2018 [2]. The released metadata contains 15,314 training, 5,055 validation, and 5,125 test cases. We use all 5,125 test cases for report generation; 5,082 have the released 30-finding classification labels.

RAD-ChestCT (external). RAD-ChestCT was used only for evaluation [40, 41]. The public release contains 3,630 volumes, each accompanied by an 84×52 abnormality-by-location label matrix extracted from its source radiology report. We evaluate a harmonized 16-label subset: 15 findings map directly, while the generic calcification target combines the CT-RATE arterial-wall and coronary-artery wall calcification outputs. The public release does not provide paired report text, so report-generation outputs are assessed through label extraction rather than reference-text similarity.

4.2. Data Curation

Our supervised fine-tuning corpus combines CT-RATE, NIH, and CancerVerse and is organized into the complementary curation families below. The resulting curriculum contains approximately 550,000 structured multimodal instruction examples across chest and abdominal regions, constructed from 70,111 unique image inputs; because one CT can support multiple tasks, the number of examples is not a count of independent examinations.

Expert CT interpretations and synthetic expansion. Working with radiologists, we collected a limited set of recorded CT interpretations. Radiologists were instructed to follow their usual image-review workflow while narrating their observations and reasoning, including provisional hypotheses, differential diagnoses, and uncertainty. They were encouraged to articulate tentative possibilities and unresolved questions, including considerations that might not be retained in a final radiology report, and to conclude with a summary of their observations. The recordings were transcribed into text, then curated and standardized into expert annotations. These annotations provided direct SFT supervision and served as exemplars for rewriting source reports into synthetic narrated interpretations that followed the organization and narrative style of the expert examples. Figure 5 illustrates this process using chest and abdominal excerpts from one training case and a separate radiologist narration exemplar.

Ontology and coverage harmonization. The source reports differ in terminology, reporting style, anatomical coverage, and available annotations. We therefore developed a radiologist-guided ontology defining a common vocabulary and labeling rules for the 30 chest and 29 abdominal findings evaluated in this work. For each abnormality, the ontology specifies accepted synonyms and aliases, inclusion and exclusion criteria, negation, uncertainty, and temporal-status rules, anatomical and severity qualifiers, overlap and parent-child relationships, and region-specific reporting conventions. A language-model-assisted labeling pass mapped the heterogeneous reports to this fixed inventory, with deterministic checks of annotation structure and consistency between parent and child labels. Chest, abdomen, and pelvis coverage was assessed separately as full, limited, not imaged, or unknown. Negative labels were used only when the relevant region was fully covered; for limited or unknown coverage, an unmentioned finding was masked rather than treated as absent. An explicitly reported positive could still support a targeted task when coverage was incomplete. The derived *No Chest Finding* and *No Abdominal Finding* labels were assigned only after full-coverage checks. Collaborating radiologists reviewed the ontology definitions, standardized schemas, and representative outputs. The complete finding inventory is listed with the per-finding evaluation in Table 9.

Structured-report generation. Eligible reports were rewritten into fixed chest- or abdomen-specific schemas. The chest schema organizes findings by medical devices, lungs and airways, pleura, mediastinum and hila, heart and pericardium, great vessels, esophagus, chest wall and breast, and bones; the abdominal schema analogously covers medical devices, hepatobiliary organs, spleen, pancreas, adrenal glands, kidneys and urinary tract, bowel, peritoneum, lymph nodes, vessels, abdominal wall and soft tissues, and bones. Both terminate in a concise impression. Rewriting retained current, image-supported findings, uncertainty, and clinically appropriate differentials while excluding history, comparison-only statements, recommendations, and unsupported content. Outputs were checked for the required headings and order, consistency of finding polarity, and agreement with the extracted ontology labels. The family contains complete explicit-region prompts, region-specific prompt variants, and generic structured-report prompts; prompt augmentation changes the instruction while preserving the same target report.

Reasoning walkthroughs. For the same eligible chest and abdominal studies, the reports were converted into continuous, first-person radiologist search-pattern walkthroughs. These examples move systematically through the relevant anatomy, connect important positive and negative observations, and retain only supported uncertainty or differential reasoning. They exclude patient-history leakage, prior-comparison claims, recommendations, dataset references, and unsupported precision. Each walkthrough ends with the canonical positive-label set, providing both free-form reasoning supervision and an auditable structured answer. Additional prompt variants change the wording of the request without altering the target reasoning.

Single-turn follow-up QA. We generated clinically natural follow-up questions covering finding presence or absence, location, differential diagnosis, management implications, clinical context, unsupported requests, terminology, and deterministically derived scan coverage or contrast questions. Each answer was constrained to the image-grounded report context and paired with concise reasoning. For the single-turn family, we retained only questions that remain unambiguous when separated from their source conversation and rejected

Constructing reasoning supervision

Radiologist exemplars guide the expansion of report findings into narrated reasoning



Figure 5 | Radiologist-guided construction of reasoning supervision. A radiologist narration from a separate case illustrates the exemplars used to guide synthetic rewriting. Chest and abdominal excerpts from one training case show original report findings, selected structured-report sections, and synthetic reasoning narrations. Blue highlights indicate observations; warm highlights indicate interpretation or uncertainty. Excerpts retain their source wording, with bracketed ellipses marking omitted passages and original numbering retained for selected impression items.

dangling references that require an earlier turn.

Multi-turn follow-up dialogue. The complementary multi-turn family preserves the initial structured-report request and four subsequent follow-up exchanges, allowing later questions to refer to findings established earlier in the conversation. Responses remain concise and report-grounded, and unsupported questions are answered by stating the evidentiary limitation rather than inventing a finding.

Binary abnormality QA. For each chest or abdominal ontology finding, we formed explicit positive and negative question pools from CT-RATE, NIH, and CancerVerse, applied per-label caps, and deterministically sampled from each source pool. Targets are direct yes/no answers with brief supporting reasoning.

Section-specific reporting. Structured reports were decomposed into their canonical anatomical sections, and ontology findings were mapped to the section in which they should appear.

Anatomical localization QA. This family comprises left/right/bilateral questions, fine anatomical-location questions, and contrastive location yes/no questions. The finding and its location had to co-occur affirmatively within the same canonical report section. Negated, resolved, ambiguous, or cross-section associations were discarded. Sampling balanced laterality within findings, capped bilateral dominance, limited each study to three questions.

Severity and attribute QA. Qualitative size, burden, multiplicity, severity, distribution, fluid amount, and extent were extracted from reports. As in localization curation, the abnormality and its descriptor had to be unambiguously bound within the same report section; a positive label alone did not license an attribute.

Refusal and input-validity supervision. Two small safety-oriented training sets were added. The first pairs valid CT volumes with unrelated or nonmedical requests and teaches the model to decline the request while redirecting toward image interpretation. The second pairs CT interpretation prompts with invalid, non-CT, or anatomically mismatched 3D inputs and teaches the model not to hallucinate a CT reading. These sets were audited to remove accidentally valid prompt-image pairs. A small set of external invalid-image examples was added for refusal supervision.

5. Results

We evaluate NV-Reason-CT on CT-RATE, Merlin, and RAD-ChestCT for abnormality classification and report generation across chest and abdominal CT. Evaluation on a held-out NIH cohort extends the assessment to a broader inventory of chest and abdominal findings. A preliminary study with expert radiologists further examines the perceived quality and usefulness of the generated reports and clinical explanations.

Figure 6 summarizes selected classification and report-generation results across the three public benchmarks. The following subsections provide the full comparisons and their evaluation protocols.

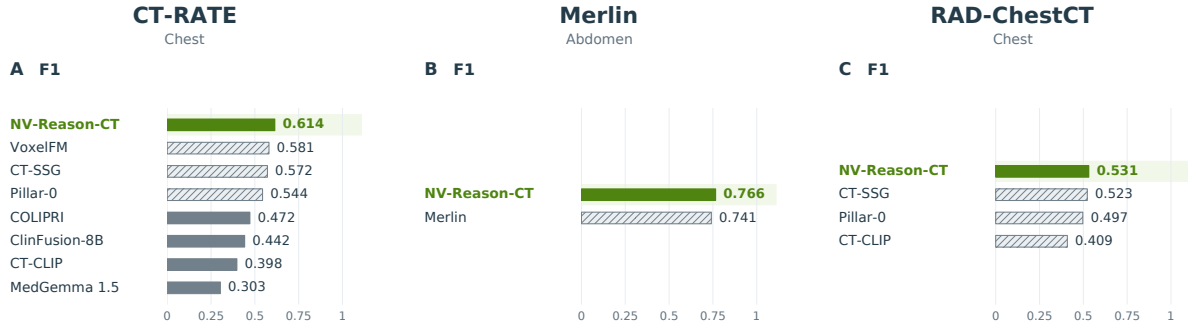
5.1. CT-RATE evaluation

Dataset composition and split construction are described in Section 4.1. Unless otherwise stated, our CT-RATE experiments use the corrected v2 validation cohort of 3,002 chest CT reconstructions, corresponding to 1,564 studies from 1,304 patients. Several published comparators instead use the historical 3,039-reconstruction manifest or evaluate one image-report pair for each of the 1,564 studies. We therefore report the cohort and evaluation unit for each result rather than treating these protocols as interchangeable. Table 2 summarizes the training split and the historical v1 and corrected v2 validation protocols. A study denotes one CT acquisition, whereas a case denotes one reconstructed 3D volume; our v2 evaluation treats each reconstruction as one example [38]. Figure 7 shows positive-case counts for the 18 abnormalities.

5.1.1. CT-RATE abnormality classification

Unified evaluation strategy. Our primary comparison target is image-only classification of all 18 original CT-RATE abnormalities for all 3,002 cases. No validation report text is supplied at inference. Our key metrics are macro-F1 score and macro-AUROC. For abnormality c , we compute the positive-class F1 and

ABNORMALITY CLASSIFICATION



REPORT GENERATION

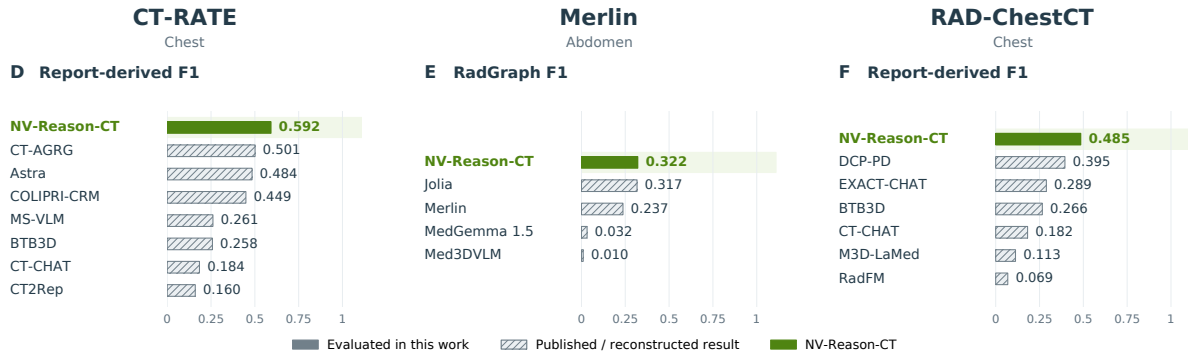


Figure 6 | Performance across CT benchmarks. Selected abnormality-classification (A–C) and report-generation (D–F) results on CT-RATE, Merlin, and RAD-ChestCT. F1 in panels A–D and F denotes positive-class macro-F1 across abnormalities; panel E reports RadGraph F1 for clinical-entity and relation overlap, using complete matching for NV-Reason-CT. Green highlights NV-Reason-CT. Solid bars denote evaluations performed in this work; hatched bars denote published or reconstructed results. Differences in cohorts, training, inference, label mapping, and evaluation are detailed in Tables 3, 4, 5, 6, 7, and 8. All scores are on a [0, 1] scale; higher is better.

Table 2 | CT-RATE cohort composition. A study denotes one CT acquisition and a case denotes one reconstruction-level 3D volume. Cases from the same study share a radiology report and its extracted labels. The original 3,039-case validation manifest was revised to 3,002 cases after 37 non-chest volumes, predominantly head CTs, were excluded.

Split / manifest	Cases	Studies	Patients
Training	47,149	24,128	20,000
Validation (v1 / v2)	3,039 / 3,002	1,564	1,304
Total (original release)	50,188	25,692	21,304

then average equally across labels,

$$\text{MacroF1}_{18} = \frac{1}{18} \sum_{c=1}^{18} \frac{2 \text{TP}_c}{2 \text{TP}_c + \text{FP}_c + \text{FN}_c}. \quad (1)$$

For our continuous-score evaluations, binary predictions use a fixed threshold of 0.5, $\hat{y}_{ic} = \mathbf{1}\{p_{ic} > 0.5\}$. Macro-AUROC is the unweighted mean of the 18 per-label AUROCs computed from the continuous scores. For evaluations based on generated Yes/No answers, F1 is computed from the parsed answers; AUROC requires a separate continuous score.

We emphasize macro-F1 because it measures the precision–recall tradeoff at the evaluated operating point, including the effect of false-positive predictions for uncommon abnormalities. AUROC provides complementary information about ranking across thresholds but does not directly describe precision and recall at the chosen operating point.

Published CT-RATE results differ in F1 averaging, operating thresholds, evaluation cohorts, and classification interfaces. Some average prevalence-weighted positive- and negative-class F1, whereas others report positive-class F1. Thresholds may be fixed, calibrated on training-derived data, or optimized on public validation labels. Native prompted inference, supervised classifier probes, ensembling, and case filtering introduce further differences. Table 3 is therefore a protocol-aware comparison rather than an undifferentiated leaderboard.

Where released checkpoints and inference code were available, we re-evaluated models on the same 3,002-case v2 manifest using the authors’ checkpoint, model-specific image preprocessing, and the closest documented native classification interface. For contrastive models, positive and negative text similarities were normalized as a two-way probability and thresholded at 0.5. No threshold tuning or calibration was applied in these re-evaluations; model-specific prompts and additional prompt comparisons are described below. Retaining each model’s native preprocessing and text encoder is necessary for a faithful reproduction, although it prevents the comparison from isolating architecture alone. Table 3 distinguishes our re-evaluations from published results, including results obtained with additional supervised classifier training. The per-abnormality F1 and AUROC results are reported in Appendix Tables A1 and A2, respectively.

Table 3 | Multi-label abnormality classification evaluation on the official public validation split. Macro-F1 is the unweighted mean of the 18 positive-class F1 scores (at 0.5 unified threshold when applicable); all metrics are reported on a $[0, 1]$ scale.

Method	Pretraining paradigm	Vision Encoder	Classification adaptation / Source	Macro-F1 $\uparrow$	Macro-AUROC $\uparrow$
CT-CLIP [1]	CLIP based image-text contrastive alignment	Factorized ViT (3D patches 20x20x10, 2D in-plane transformer, followed by out-of-plane, resample 0.75x0.75x1.5mm, crop 480x480x240)	Our eval using <i>Zero-shot</i> official checkpoint. Pos+Neg prompts.	0.3976	0.7328
			Our eval using <i>VocabFine</i> official checkpoint. Pos+Neg prompts.	0.2662	0.7583
			VoxelFM [15] eval: retrained extra head classifier.	0.2760	0.5740
Merlin [2]	CLIP based image-text contrastive alignment	3D ConvNet, inflated from ResNet152, (224x224x160 crop, 1.5x1.5x3mm resampling)	Our eval using official checkpoint. Pos+Neg prompts.	0.3579	0.6617
			VoxelFM [15] eval: retrained extra head classifier.	0.4955	0.7983
COLIPRI [3]	Multi-objective: CLIP based, VQA, image-only SSL pretraining	3D ViT (8x8x8 patch, 192x192x192 voxels input, 2x2x2mm resampling)	Authors' supervised-probe AUROC; our F1 eval using the released checkpoint and Pos+Neg prompts.	0.4717	0.8615
Jolia [16]	CLIP based image-text contrastive alignment	3D ConvNet stem + transformer (192x192x192 crop, 1.5x1.5x1.5mm resampling, 11 CT-window channels)	Our eval using official checkpoint. Contrastive [CLS]-only zero-shot using 8 Pos+Neg prompt pairs.	0.3589	0.6986
Pillar-0 (Chest) [14]	CLIP based image-text contrastive alignment	3D ConvNet stem+transformer (256x256x256 crop, 1.25x1.25x1.25mm resampling, 11 CT-window channels)	DCP-1 eval [24]: retrained extra head classifier.	0.5440	0.8610
VoxelFM [15]	DINO based image only pretraining	3D ViT (14x14x14 patch, 112x112x122 voxels input, isotropic resample to smallest spacing)	Authors' eval: single-layer Q-Former classifier on top of the vision encoder.	0.5807	0.8701
CT-SSG [17]	ResNet18 init	2.5D ConvNet + graph. crop 480x480x240, 1.5x1.5x0.75 resampling.	Authors' eval	0.5718	0.8364
RadFM [29]	VLM instruction tuning	3D ViT + Perceiver (32x32x4 patch size, 64-slice input)	Our eval using the official checkpoint. Native generated Yes/No answers.	0.2153	–
			VoxelFM [15] eval: retrained extra head classifier.	0.4325	0.7592
M3D-LaMed [30]	Multi-objective: CLIP based, VQA	3D ViT (16x16x4 patch size, resize input to 256x256x32)	Our eval using the official checkpoint. One-token Yes/No forced choice.	0.3046	–
			VoxelFM [15] eval: retrained extra head classifier.	0.4257	0.7403
MedGemma 1.5 [31]	Frozen MedSigLIP vision encoder; VQA	2D ViT (sequence of up to 85 axial 480x480 slices, 0.75x0.75x1.5 mm resampling)	Our eval using official checkpoint. Yes/No prompt per abnormality.	0.3033	–
ClinFusion [32]	2D/3D VLM instruction tuning	2D and 3D ViT fusion (256x256x32 crop, 8x8x8 patch)	ClinFusion-8B checkpoint. Single prompt for Yes/No.	0.4419	–
			ClinFusion-32B checkpoint. Single prompt for Yes/No.	0.3741	–
NV-REASON-CT (ours)	VLM instruction tuning: VQA, reports, reasoning. SFT + GRPO/RL	3D ViT (8x8x8 patch, 192x192x192 voxels input, 2x2x2mm resampling)	Single prompt for Yes/No abnormality presence (uniform template) for F1 score. Yes/No token probability for AUROC.	0.6140	0.8710

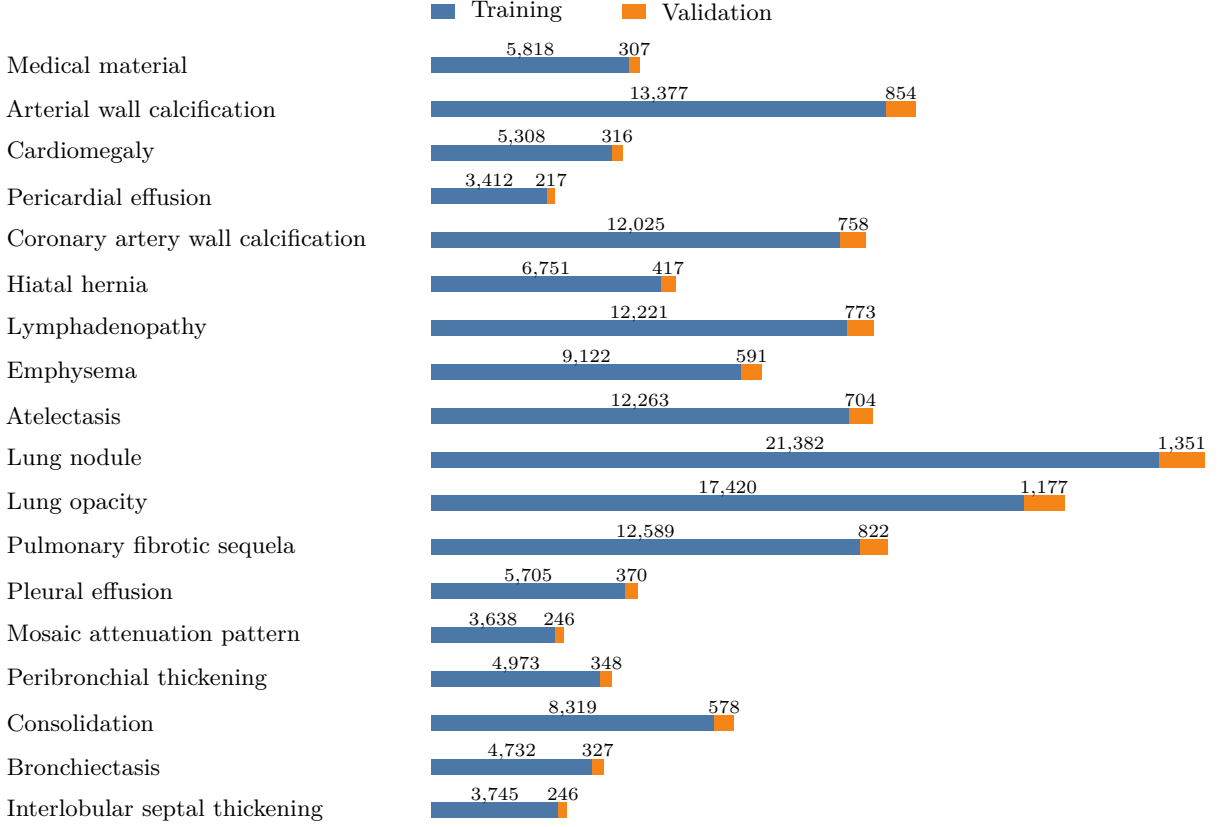


Figure 7 | CT-RATE class (abnormality) composition in the training and v2 validation cohorts. Horizontal bars show the number of positive cases for each of the 18 abnormality labels; Labels are not mutually exclusive, several abnormalities can occur in the same CT case.

NV-Reason-CT. NV-Reason-CT achieved macro-F1 0.6140 using generated binary answers and macro-AUROC 0.8710 using a separate token-probability evaluation. For macro-F1 evaluation, we used the uniform binary prompt “{finding}: present or absent in this CT? Answer Yes (for present) or No (for absent).” We parsed the Yes or No text output after the generated thinking.

For AUROC, we disabled thinking and applied a two-way softmax to the Yes and No logits at the first generated position, using the same binary prompt. The resulting Yes probability provided the continuous score. Thus, the reported F1 measures generated-answer behavior, whereas AUROC measures ranking in the separate probability-scoring mode.

Additional uniform binary prompts produced similar results. We also evaluated a single prompt requesting all present abnormalities from the allowed label inventory, with “No Chest Finding” returned when none was present. This yielded macro-F1 0.6100, compared with 0.6140 for individual abnormality questions, while requiring only one inference call per CT.

CT-CLIP. CT-CLIP [1] was released jointly with CT-RATE as a baseline for the dataset. CT-CLIP aligns volumetric CT and report representations using a factorized ViT vision encoder and a CXR-BERT-family text encoder; its VocabFine variant additionally adapts the text vocabulary to CT terminology. The original publication reports mean F1 values of 0.7069 and 0.7384 and macro-AUROC of 0.7308 and 0.7561 for the zero-shot and VocabFine checkpoints, respectively. These F1 values average prevalence-weighted binary F1 and use per-label operating points, so they are not directly comparable to the fixed-threshold positive-class macro-F1 used here.

We re-evaluated both released checkpoints on the 3,002-case v2 cohort with the authors’ volume pre-processing. Each label was scored with “{abnormality} is present.” and “{abnormality} is not present.”; a two-way softmax over image–text similarities defined $P(\text{present})$, with no calibration or prompt variation. We

used the prompt formulation selected and released by the CT-CLIP authors. This yielded macro-F1/AUROC of 0.3976/0.7328 for zero-shot and 0.2662/0.7583 for VocabFine. Separately, VoxelFM [15] trained a classifier on frozen CT-CLIP features and reported macro-F1/AUROC of 0.2760/0.5740 on the historical 3,039-case cohort. Table 3 includes this supervised-probe result as a distinct evaluation.

Merlin. We evaluated the official Merlin checkpoint [2], with its released preprocessing, on all 3,002 CT-RATE v2 cases. L2-normalized image and text embeddings were compared using the uniform pair “{finding} is present.” and “No {finding}.”, and the learned logit scale and a two-way softmax produced the positive probability. At the fixed 0.5 threshold, this native contrastive evaluation obtained macro-F1/macro-AUROC of 0.3579/0.6617.

Using the CT-CLIP prompt pair, “{abnormality} is present.” and “{abnormality} is not present.”, yielded lower macro-F1/macro-AUROC of 0.3081/0.5931. Independently, the VoxelFM study trained a supervised classifier on the frozen Merlin visual encoder, obtaining macro-F1/macro-AUROC of 0.4955/0.7983 on 3,039 reconstructions at a fixed 0.5 threshold [15].

COLIPRI. The COLIPRI publication [3] reports macro-AUROC 0.8615 on CT-RATE using a supervised classification probe, but does not report macro-F1. We re-evaluated the released COLIPRI-CRM checkpoint in its native contrastive mode on all 3,002 v2 cases. The official processor resampled volumes to 2-mm isotropic spacing and cropped or padded them to $192 \times 192 \times 192$ voxels. For each label, we used “No {abnormality} present” and “{abnormality} present”, divided the two image-text similarities by the released temperature, and applied a two-way softmax. We used the prompt formulation selected by the COLIPRI authors. No threshold tuning or calibration was performed. This evaluation produced macro-F1 0.4717 and macro-AUROC 0.8161 and required no additional CT-RATE-supervised training.

Table 3 reports our contrastive macro-F1 alongside the authors’ supervised-probe macro-AUROC; these values come from different evaluation configurations.

Pillar-0. The Pillar-0 Chest result [14] comes from the separate DCP-PD study, which freezes the Atlas visual encoder and trains a linear multi-label classifier on CT-RATE training data. This classifier is denoted DCP-1 [24]. The reported all-label averages are macro-F1 0.544 and macro-AUROC 0.861 on the historical 3,039-case validation manifest. This entry therefore evaluates supervised classification on frozen Pillar-0 features.

Jolia. Jolia [16] reports a short-prompt, [CLS]-only macro-AUROC of 0.7346 using 1,564 acquisition-level examples; a directly comparable positive-class macro-F1 is not reported. Our evaluation used the released checkpoint, but scored all 3,002 images. We used the global [CLS] embedding and the eight positive/negative prompt pairs listed in the publication, averaged cosine similarity within the positive and negative prompt sets, and applied a two-way softmax with a fixed 0.5 threshold. We did not use organ-query features, calibration, or a trained linear probe. The resulting reconstruction-level macro-F1 was 0.3589 and macro-AUROC was 0.6986. The difference from the reported AUROC should not be attributed to model variation alone because the evaluation unit and manifest also differ.

VoxelFM. VoxelFM [15] is an image-only volumetric model pretrained with a DINO-style self-supervised objective. For CT-RATE, the authors train a single-layer Q-Former classifier using the official training labels, reserving 20% of the training split for internal validation, and evaluate all 18 abnormalities on the historical 3,039-case validation set. The reported fixed-0.5 result is macro-F1 0.5807 and macro-AUROC 0.8701. The study provides a complete per-label breakdown and applies the same head and scoring protocol to several comparator encoders, making it a useful supervised-probe reference. We use the authors’ reported result without modification.

CT-SSG. CT-SSG uses an ImageNet-initialized ResNet-18 to encode adjacent-slice triplets, followed by graph-based spectral aggregation for multilabel CT classification [17]. Its five-fold evaluation reports macro-F1 0.5718 and macro-AUROC 0.8364 on the historical public validation cohort, with all 18 per-label values available.

RadFM. We evaluated the public RadFM checkpoint [29] following the released inference example, querying each abnormality with “Can you identify any visible signs of {abnormality} in the image?”. The released volume

loader was adapted to CT-RATE’s array orientation before resizing axial slices to 512×512 , resizing depth to 64, replicating the grayscale volume across three channels, and applying per-volume min–max normalization; no HU windowing or physical-spacing resampling was used. Native generation yielded positive-class macro-F1 0.2153.

Separately, VoxelFM discarded the language-model component and trained a supervised classifier on the frozen RadFM visual encoder, obtaining macro-F1 0.4325 and macro-AUROC 0.7592 on the historical 3,039-case cohort [15]. The two rows in Table 3 therefore measure native prompted behavior and supervised representation transfer, respectively.

M3D-LaMed. We evaluated the public M3D-LaMed Phi-3-4B checkpoint [30]. Each CT image was transposed to depth-first order, min–max normalized, foreground-cropped, and resized to $32 \times 256 \times 256$. This is an independent raw-volume adaptation because M3D-LaMed was trained with rendered 8-bit slice stacks and its release does not define raw-HU CT-RATE windowing. We used “Is {abnormality} present in this CT scan? Answer only Yes or No.” This evaluation yielded macro-F1 0.3046. Separately, VoxelFM trained a supervised classifier on the frozen M3D visual encoder, without the language model, and reported macro-F1 0.4257 and macro-AUROC 0.7403 on the historical 3,039-case cohort [15].

MedGemma 1.5. MedGemma 1.5 is a 4B medical 2D VLM with a MedSigLIP visual encoder that accepts a sequence of sampled 2D CT slices rather than a native 3D image [31]. The technical report evaluates up to 85 axial slices per scan and reports CT-RATE macro-F1 values from 0.235 to 0.269 on 1,558 acquisition-level examples.

Our evaluation uses the official checkpoint, the report’s three-window slice construction, $0.75 \times 0.75 \times 1.5$ -mm resampling, and the published per-abnormality prompt: “You are an expert radiologist for chest CT. Looking at these CT slices, is there {abnormality}? Answer with ‘Final Answer: yes’ or ‘Final Answer: no.’” This generated-answer evaluation yielded macro-F1 0.3033. AUROC was not computed from the discrete answers.

ClinFusion-8B and ClinFusion-32B. We independently evaluated both official checkpoints: ClinFusion-8B built on Qwen3-VL-8B-Instruct, and ClinFusion-32B built on Qwen3-VL-32B-Instruct [32]. Both evaluations covered all 18 labels and all 3,002 CT-RATE validation cases. We used 1-mm isotropic resampling, resizing to $32 \times 256 \times 256$, and deterministic selection of four 2D anchor slices; both the 3D and 2D visual streams were retained. Each abnormality was evaluated independently with “Is {label} present in this CT scan? Answer only Yes or No.” Macro-F1 was 0.4419 for ClinFusion-8B and 0.3741 for ClinFusion-32B. These results reflect our independently implemented evaluation and prompt selection; alternative protocols or prompt formulations may yield different results.

Notable results excluded from the direct comparison. AnyMC3D [42] reports macro-F1 0.646 with class-specific calibration for a ten-model ensemble learned from a separate training-derived holdout; the same ensemble obtained 0.426 without calibration. We do not include the calibrated ensemble in Table 3 because it changes both model multiplicity and class-specific operating points relative to the single-model evaluations emphasized here. Other non-matched results in the table are retained only with the explicit protocol qualifications above.

fVLM [18] is also not directly included because its released evaluator requires corrected-HU volumes and TotalSegmentator-derived masks for four anatomical regions. It provides organ-conditioned zero-shot scores for only 16 of the 18 CT-RATE labels.

5.1.2. CT-RATE report generation

We evaluate chest CT report generation on the public CT-RATE validation cohort described in Section 5.1. Our unified evaluation uses all 3,002 chest reconstructions in the corrected v2 cohort. NV-REASON-CT generates the Findings and Impression sections without access to the reference report. Clinical-content metrics are computed from the 18 CT-RATE abnormalities, whereas text metrics compare the generated sections with the corresponding reference sections. Because several reconstructions can originate from the same study, they share one acquisition-level reference report. Published comparators variously evaluate the historical

3,039 reconstructions or 1,564 study-report pairs; we therefore state the evaluation unit and cohort for every result in Table 4.

We organize the reported metrics into clinical-content and report-similarity measures. For report-derived Macro-F1 metric, the official CT-RATE RadBERT labeler maps each generated report to 18 abnormality probabilities, which are thresholded at 0.5 and compared with the reference labels. Positive-class F1 is computed independently for each abnormality and averaged with equal weight across all 18 labels. CRG metric uses the resulting true-positive, false-negative, and false-positive counts but excludes true negatives; it applies prevalence-dependent rewards and penalties before normalization, reducing the inflation produced by class imbalance and trivial predominantly normal reports [20]. GREEN uses a learned evaluator to identify clinically relevant errors relative to a reference report. BLEU-4 measures four-gram precision with a brevity penalty, ROUGE-L measures longest-common- subsequence overlap, and METEOR aligns unigrams while allowing stemming and synonym matches. All metrics in Table 4 are reported on a $[0, 1]$ scale, with higher values indicating better performance.

We treat report-derived Macro-F1 and CRG as the primary clinically oriented metrics in this evaluation. Both directly test whether abnormalities are correctly communicated and penalize missed findings and unsupported findings; Macro-F1 gives rare and common abnormalities equal weight, while CRG further accounts for label prevalence and is not increased by abundant true negatives. The natural language metrics BLEU-4, ROUGE-L, and METEOR primarily reward n-gram or sequence agreement with a single report and are therefore sensitive to clinically inconsequential differences in terminology, sentence order, section organization, and report length. This consideration is particularly relevant because NV-REASON-CT produces standardized Findings and Impression sections, whereas the CT-RATE references retain heterogeneous narrative styles. Prior CT report-generation studies have similarly noted that natural-language metrics may fail to recognize semantically equivalent phrasing or fully capture diagnostic fidelity [22, 1, 20, 43]; GREEN improves on purely lexical comparison but remains dependent on a reference report and a learned evaluator. Its official checkpoint was fine-tuned primarily on chest X-ray reports, making its application to 3D CT reports out of domain and potentially less reliable [20]. We report all metrics for comparability while placing greater interpretive weight on clinically grounded abnormality metrics.

Table 4 | CT-RATE radiology report-generation results. Unless otherwise noted, F1 denotes positive-class macro-F1 over the 18 CT-RATE abnormalities extracted from generated reports. All scores are reported on a $[0, 1]$ scale.

Method	Evaluation details	F1 $\uparrow$	CRG $\uparrow$	GREEN $\uparrow$	BLEU-4 $\uparrow$	ROUGE-L $\uparrow$	METEOR $\uparrow$
CT2Rep [19, 20]	Authors’ evaluation on 3,039 reconstructions	0.160	0.359	0.487	0.213	0.362	0.197
CT-CHAT [1, 20]	Authors’ evaluation on 3,039 reconstructions	0.184	0.368	0.436	0.198	0.326	0.215
BTB3D [21]	Authors’ evaluation on 3,039 reconstructions	0.258	0.370	–	0.215	–	0.223
RadFM [29, 20]	CRG evaluation [20]; official pretrained checkpoint	0.059	0.335	0.018	0.000	0.042	0.020
CT-SSG [17]	Authors’ evaluation on the public CT-RATE validation partition; five-fold mean; F1 averaging not specified	0.3872	0.4366	–	0.107	0.246	0.164
MS-VLM [22]	Authors’ evaluation on 3,039 reconstructions	0.261	–	–	0.232	0.438	0.396
M3D-LaMed [30]	CT-Agent [44] evaluation; micro-F1	0.148	–	–	0.245	0.400	0.326
CT-AGRG [23]	Authors’ evaluation on 3,039 reconstructions; CT-Net variant	0.5011	–	–	0.172	0.280	0.196
RMR [28]	1,564 test reports; reconstruction selection or aggregation not specified	0.495	–	–	0.203	0.280	0.235
MonteRET [26]	Authors’ evaluation on 1,564 scans; macro-F1 derived from reported per-finding F1 values; reconstruction mapping not specified	0.365	–	–	0.252	0.379	0.454
Astra [27]	Authors’ evaluation on 1,564 acquisition-level cases with harmonized references; macro-F1 approximated from label-wise F1 bars in Fig. 7e [27]	0.484	–	–	0.2501	0.4409	0.2397
COLIPRI-CRM [3]	Evaluation cohort size not specified	0.449	–	–	–	–	–
VoxelFM [15]	3,039 reconstructions; reformatted reference reports and GPT-OSS-120B abnormality extraction	0.432	–	–	–	–	–
ClinFusion-8B [32]	3,039 reconstructions; reference-derived clinical context; GPT-4.1 claim-level F1	0.216	–	–	–	–	–
ClinFusion-32B [32]	3,039 reconstructions; reference-derived clinical context; GPT-4.1 claim-level F1	0.239	–	–	–	–	–
NV-REASON-CT (ours)	Our evaluation on 3,002 v2 reconstructions; RadBERT extraction from generated Findings and Impression	0.5923	0.5138	0.40722	0.1145	0.2140	0.1921

NV-REASON-CT. NV-REASON-CT is our native 3D generative VLM, trained with supervised fine-tuning and reinforcement learning on VQA, structured-report, and reasoning tasks.

The report evaluation compares generated Findings and Impression with the corresponding reference sections for all 3,002 reconstructions (1,564 studies from 1,304 patients). Generated thinking text, technique, and the auxiliary structured label list are excluded from report scoring. We apply the official CT-RATE RadBERT label extractor with a fixed 0.5 sigmoid threshold to obtain the report-derived macro-F1 of 0.5923 and use the same extracted labels for CRG. BLEU-4, ROUGE-L, and METEOR are computed with the COCO-caption implementations, and GREEN is computed with its official evaluator. The two F1 values therefore measure distinct outputs and should not be treated as interchangeable.

Our generated reports end with a summary list of detected abnormalities, including findings from the 18-label CT-RATE inventory. This list allows direct computation of F1 without the RadBERT extractor. Using such direct computation, our model achieves 0.6010 macro-F1 score from the report. In Table 4 we decided to report the RadBERT report derived macro-F1 of 0.5923 for consistency with other methods.

Compared to binary QA per-abnormality prompt F1 score of 0.6140 in Table 3 the report based F1 score of 0.6010 is slightly lower, but includes richer information (location and severity of abnormalities, or abnormalities beyond 18 ct-rate labels).

CT2Rep. CT2Rep [19] is an early 3D sequence-to-sequence report generator trained from scratch on CT-RATE findings. Its visual encoder operates on volumetric patches, and a Transformer decoder combines relational memory with memory-conditioned normalization to generate report text. It comes from the authors of CT-RATE dataset, which they later evaluated on report generation in CRG study of the released model on the historical 3,039-reconstruction validation cohort, using an 18-label report classifier and standardized text metrics [20].

CT-CHAT. CT-CHAT [20] connects the frozen CT-CLIP volumetric encoder to Llama 3.1 through a learned projector [1]. It first aligns CT features to report text and then applies visual instruction tuning with low-rank

language model adaptation; The report benchmark uses the 8B language-model configuration and the historical 3,039 reconstruction-level validation examples.

BTB3D. BTB3D learns discrete volumetric tokens with a causal 3D convolutional encoder–decoder, Haar-wavelet compression, and lookup-free quantization, then maps those tokens to a Llama 3.1-8B report decoder [21]. We use the best reported $16 \times 16 \times 8$ tokenization result. Its test set is the historical public cohort of 3,039 images.

RadFM. RadFM is a generalist medical VLM comprising a 3D ViT, a Perceiver-style token resampler, and a MedLLaMA-13B LLM [29]. CRG evaluates the official pretrained model directly on the historical 3,039 CT-RATE reconstructions, without CT-RATE-specific report retraining [20].

MS-VLM. MS-VLM combines a self-supervised 2D DINO slice encoder with a Z-former for inter-slice modeling and a learned bridge to a largely frozen language model [22]. After representation alignment, it is jointly tuned for report generation and VQA. It’s evaluated on 3,039 images.

M3D-LaMed. M3D-LaMed pairs a native 3D ViT with a Llama-family language model and is trained through image–text contrastive alignment followed by multimodal instruction tuning on the M3D-Data corpus [30]. Its original publication evaluates report generation on M3D-Cap rather than CT-RATE. The CT-RATE values in Table 4 come only from the CT-Agent comparison on the stated 3,039-case cohort and use CT-Agent’s micro-averaged clinical-efficacy protocol [44].

CT-AGRQ. CT-AGRQ first trains a CT-Net or CT-ViT multilabel classifier, then uses a PubMed-pretrained GPT-2 decoder to generate one sentence for each predicted positive abnormality and concatenates the sentences into a report [23]. The classifier uses abnormality-specific thresholds selected on an internal validation split, so report content is explicitly gated by supervised label decisions rather than generated as a single open-ended narrative. The reported CT-Net result is the mean of five runs on 3,039 images.

COLIPRI-CRM. COLIPRI pretrains a 3D ViT using image–report contrastive learning, report generation, opposite-sentence discrimination, and image-only self-supervision [3]. For its CRM report experiment, the visual encoder is frozen and a two-layer projector and Qwen2.5-1B decoder are trained on CT-RATE findings with next-token supervision. The authors report RadBERT macro-F1 0.449 on a withheld CT-RATE test set, but do not state its number of images. The cohort equivalence to either 3,039 reconstructions or 1,564 report-level studies remains unknown.

Resolution Meets Reduction (RMR). RMR [28] uses a Curia-64S slice encoder, a Perceiver resampler, and Gemma3-4B; the vision encoder is frozen while the projector and language-model LoRA weights are optimized with next-token report supervision. The evaluation is on 1,564 official validation reports. Evaluation is therefore acquisition/report-level, whereas the 3,039 protocol is reconstruction-level. The exact image selection (or aggregation) is not specified.

MonterET. MonterET combines report generator with condition- and region-aware retrieval, reranking, and LLM refinement [26]. It is tested on 1,564 scans. The authors report RadBERT micro-F1 0.420; the 0.365 entry in Table 4 is our unweighted mean of the 18 per-abnormality F1 values printed to three decimals. It is a derived approximation, and neither the scan-to-reconstruction mapping nor the retrieval database available to each test case is specified.

Astra. Astra connects a Merlin visual encoder through a Perceiver to Qwen2.5-VL and is trained using supervised fine-tuning followed by FORTE-guided reinforcement learning [27]. Its CT-RATE evaluation uses 1,564 acquisition-level cases and harmonized reference findings, rather than the native reports paired independently with every released reconstruction. The authors report RadBERT micro-F1 0.5502; the approximately 0.484 macro-F1 in our table is an unweighted mean digitized from the 18 label-wise bars in their Fig. 7e. Both digitization uncertainty and the harmonized references preclude treating this as an exact, protocol-matched macro-F1.

VoxelFM. VoxelFM is a 3D ViT pretrained with DINO-style self-supervision [15]. For report generation, the encoder is frozen and a Q-Former interface to Qwen3-8B is trained on CT-RATE with low-rank language-model

adaptation. The authors reformat reference reports with GPT-OSS-120B and use the same model to extract the 18 findings from generated reports. They evaluate the full historical public validation cohort, reported as 3,039 reconstructions.

CT-SSG. CT-SSG is a 2.5D system that combines an ImageNet-initialized ResNet-18 over adjacent slices with graph-based spectral aggregation [17]. For reporting, it freezes the classification-pretrained visual encoder and trains a lightweight CT2Rep-style autoregressive decoder with next-token supervision. Its results are reported as a five-fold CT-RATE evaluation with RadBERT label extraction. The paper states the evaluation is on the CT-RATE test cohort, but the number of report-generation test examples is not specified.

ClinFusion. ClinFusion augments Qwen3-VL with Qwen-ViT, DINOv2-Large, ConvNeXt-Large, and a native 3D ViT fused through a cascade spatial-aware locality module [32]. The 8B and 32B variants are trained through five alignment and instruction-tuning stages. Both variants are evaluated on 3,039 historical CT-RATE cases, but generation is conditioned on clinical indication and anatomical regions extracted from the reference report by GPT-4.1. GPT-4.1 then labels generated claims as matched, missed, or hallucinated, giving claim-level F1 of 0.216 and 0.239.

Notable method not evaluated. Merlin is a 3D vision–language foundation model pretrained on abdominal and abdominopelvic CT examinations and their corresponding radiology reports [2]. Its adapted report-generation module was likewise trained for section-by-section abdominal reporting and evaluated on the authors’ internal abdominal CT test set, not on chest CT. Although the released generator includes a lower-thorax section, reflecting the lung bases visible on abdominal examinations, it does not provide a native complete chest-report protocol. We therefore exclude the official Merlin checkpoint from Table 4.

5.2. Merlin dataset evaluation

The Merlin Abdominal CT Dataset comprises 25,494 abdominal and abdominopelvic CT scan–report pairs from 18,317 patients treated at Stanford Hospital between 2012 and 2018 [2]. Each case corresponds to one CT examination, for which the DICOM series with the largest number of slices was retained and converted into a 3D NIfTI volume. The original study describes more than six million constituent two-dimensional images in its 15,331-scan training cohort. It reports patient-disjoint internal splits of 15,331 training, 5,060 validation, and 5,137 test scans. The subsequently released metadata differs slightly, containing 15,314 training, 5,055 validation, and 5,125 test scans. We use the released test split throughout. For abnormality classification, 5,082 of the 5,125 test scans have released 30-finding labels; the remaining 43 scans have reports but no row in the classification-label file. For report generation, we use the paired reports for all 5,125 released test scans. Classification uses our general checkpoint without Merlin-specific fine-tuning, whereas report generation uses the adapted checkpoint described in Section 5.2.2.

5.2.1. *Merlin abnormality classification*

We evaluate abnormality classification on the held-out Merlin abdominal CT cohort [2]. This benchmark complements CT-RATE by testing the model on a different anatomical domain and label ontology. The released labels cover 30 findings and are report-derived and tri-state: positive, explicitly negative, or unmentioned. Following the original zero-shot protocol, unmentioned findings are excluded and positive and negative examples are balanced independently for each finding before computing positive-class F1. Table 5 presents preliminary results. The comparison is not strictly paired because the original paper used an internal 5,137-CT test split and did not release its exact balanced case IDs, whereas our evaluation reconstructs balanced cohorts from the released test labels. In addition, our reported prompt was selected after prompt exploration on this same released test cohort and should therefore be regarded as a test-prompt-tuned estimate. For our probability-based evaluation, thinking is disabled and the normalized next-token probability of *Yes* is retained as the continuous score. F1 uses a fixed threshold of 0.5, without threshold tuning, whereas AUROC is threshold-free. For six lower-thoracic findings— aortic valve calcification, coronary calcification, cardiomegaly, pleural effusion, atelectasis, and hiatal hernia—we used the chest anatomical region; the remaining 24 findings used the abdomen region. The same prompt with thinking enabled and generated hard Yes/No answers yielded macro-F1 0.7531. Per-finding results for the probability-based evaluation are reported in Appendix C.

Table 5 | Merlin 30-finding abnormality-classification results. Macro-F1 is the unweighted mean of the positive-class F1 scores across the 30 findings after balancing explicit positive and negative examples separately for each finding. Merlin produces hard decisions and does not report AUROC for this task. For NV-REASON-CT, F1 is computed from next-token probabilities at a fixed 0.5 threshold and AUROC from the corresponding continuous scores. All scores are reported on a $[0, 1]$ scale.

Method	Evaluation details	Macro F1 $\uparrow$	Macro AUROC $\uparrow$
Merlin [2]	Authors’ internal 5,137-CT test split; per-finding balanced cohorts; hard comparison of mean image–text cosine similarity across radiologist-developed positive and negative prompt banks.	0.7410	–
VoxelFM [15]	VoxelFM-study evaluation on 5,082 labeled studies from the public Merlin test split; all 30 findings, with unmentioned labels treated as negative; frozen VoxelFM encoder with a supervised Q-Former probe trained on the Merlin training split and selected by validation AUROC.	–	0.7970
NV-REASON-CT (ours)	Released test labels; balanced cohort of 11,348 study–finding pairs from 4,670 CTs across 30 findings; one uniform prompt; normalized next-token Yes/No probabilities with thinking disabled; fixed 0.5 F1 threshold and no threshold tuning; chest anatomical region for six lower-thoracic findings and abdomen region for the remaining 24.	0.7655	0.8329

5.2.2. Merlin report generation

We evaluate abdominal CT report generation on all 5,125 scans in the released Merlin test split using the Findings section. We compute BLEU, ROUGE-L, BERTScore, RadGraph-F1, and GREEN using the RadEval framework [45]. BLEU measures lexical n -gram precision with a brevity penalty, ROUGE-L measures longest common subsequence overlap, and BERTScore measures contextual token-level semantic similarity. RadGraph-F1 first maps each candidate and reference report to a graph of clinical entities and their relations, then measures graph overlap with an F1 score [46, 47]. We report RadEval’s stricter complete-match RadGraph-F1. GREEN uses a learned evaluator to identify clinically significant errors relative to the reference report [48]. We follow the evaluation setup used by Jolia [16].

The released Merlin reports use a specific structure comprising short, organ-specific subsections [2]. By contrast, our native NV-REASON-CT reporting format is different: it is longer with different subsections and organizes Findings and Impression differently. Lexical and semantic similarity metrics are sensitive to report length, wording, section order, and headings; evaluating the native output directly would therefore conflate clinical content with output-format mismatch. To reduce this mismatch, we performed supervised in-domain output-format adaptation starting from our general checkpoint. We fine-tuned on the 15,314 released Merlin training rows for one epoch, without mixing other datasets. A dedicated prompt requested only the Findings section and specified the fixed order of the 15 Merlin anatomical headings. The same prompt was used to generate Findings for all 5,125 test scans; no test report was used for parameter updates.

Table 6 compares our adapted model with the author-reported results compiled by Jolia [16]. The nominal cohort and report scope are matched, although exact evaluator versions and reference preprocessing may differ.

Table 6 | Report generation evaluation on the Merlin dataset test split ($N = 5,125$).

Method	BLEU $\uparrow$	ROUGE-L $\uparrow$	BERTScore $\uparrow$	RadGraph-F1 $\uparrow$	GREEN $\uparrow$
Med3DVLM	0.004	0.079	0.256	0.010	0.007
MedGemma 1.5	0.124	0.106	0.361	0.032	0.038
Merlin	0.063	0.285	0.551	0.237	0.316
Jolia	0.119	0.323	0.567	0.317	0.324
NV-REASON-CT (ours)	0.0769	0.3360	0.5788	0.3216	0.3658

Following adaptation to Merlin’s Findings-section format, NV-REASON-CT achieves the highest reported ROUGE-L, BERTScore, RadGraph-F1, and GREEN scores among the models in Table 6. Its BLEU score is lower than those of MedGemma 1.5 and Jolia, showing that the relative ranking depends on the evaluation metric.

5.3. RAD-ChestCT dataset evaluation

RAD-ChestCT [40, 41] is an external, out-of-distribution chest CT benchmark that we use only for evaluation. The public image release contains 3,630 volumes. Each publicly released volume is accompanied by an 84×52 abnormality-by-location label matrix extracted from its source radiology report. Following prior work, we evaluate a subset of 16 harmonized abnormality labels for consistency of comparisons [1, 49]. This mapping starts from the 18-label CT-RATE abnormality ontology, excludes Mosaic attenuation, and merges the Arterial wall calcification and Coronary artery wall calcification outputs into a single RAD-ChestCT Calcification target. The remaining 15 abnormalities are matched directly, yielding 16 scan-level targets in total. Some studies use alternative 16-label mappings, which we explicitly identify below. Table 7 summarizes published direct classification results.

5.3.1. RAD-ChestCT abnormality classification

Table 7 | Published RAD-ChestCT abnormality-classification results. Macro F1 is the unweighted mean of the per-abnormality positive-class F1 scores, whereas weighted F1 averages the positive- and negative-class F1 within each abnormality according to class support before averaging across abnormalities.

Method	Evaluation details	Macro F1 ↑	Weighted F1 ↑	AUROC ↑	Precision ↑	Recall ↑
CT-CLIP [1]	3,630 volumes; 16 classes. VocabFine version. macro-F1 was approximated from the authors’ normalized per-class confusion matrices.	0.4087	0.677	0.650	0.346	0.5996
Pillar-0 [14]	3,630 volumes; 16 classes. DCP-PD [24] evaluation over frozen Pillar-0 features. Weighted binary F1 was recomputed from the authors’ per-class precision and recall values.	0.497	0.6942	0.796	0.399	0.811
CT-SSG [17]	1,334 patients (but not stated if all 3,630 volumes used); 16 classes. Five-fold mean.	0.5225	–	0.7458	–	–
fVLM [18]	3,630 volumes; 16 classes (alternative definition). Zero-shot, organ-conditioned paired-prompt inference.	–	0.688	0.680	0.374	0.646
Merlin [2]	3,630 volumes; 16 classes (alternative definition). fVLM-paper evaluation [18] under the organ-mapped set.	–	0.663	0.644	0.348	0.610
COLIPRI-CRM[3]	3,630 volumes; 16 classes (mapping not stated); best of five pooling schemes.	–	–	0.7657	–	–
NV-REASON-CT (ours)	3,630 volumes; 16 classes. Direct next-token Yes/No evaluation with a fixed $P(\text{Yes}) \geq 0.5$ rule, no re-training, no calibration.	0.5311	0.7379	0.7958	0.4503	0.7647

NV-REASON-CT. We evaluated all 3,630 public RAD-ChestCT volumes using the 16-label mapping described above, a single binary prompt, and a fixed 0.5 decision threshold, with no RAD-ChestCT-specific prompt or threshold tuning. Direct classification achieved positive-class macro-F1 0.5311, weighted binary F1 0.7379, and macro-AUROC 0.7958.

CT-CLIP. CT-CLIP was contrastively pretrained on the same CT-RATE training split of 47,149 reconstructed volumes and their reports. It was evaluated on 3,630 RAD-ChestCT volumes using the 16-label harmonization described above.

fVLM. fVLM learns organ-conditioned image–text representations using anatomical masks and one learned query per organ [18]. The authors evaluate the whole 3,630-volume public RAD-ChestCT release with paired positive/negative prompts. Its released evaluator implements an alternative 16-label mapping and requires TotalSegmentator-derived masks.

Merlin. The official Merlin encoder was pretrained on 15,331 abdominal CT scans with reports and structured clinical data [2]. The fVLM paper [18] evaluates Merlin on all 3,630 RAD-ChestCT volumes using this alternative 16-label mapping; we report these results in Table 7.

COLIPRI. COLIPRI [3] combines image–text contrastive learning, report generation, opposite-sentence learning, and image-only self-supervision. The authors’ frozen probe is trained and selected only with CT-

RATE data, sweeping five pooling schemes and four learning rates. It uses all 3,630 RAD-ChestCT volumes; the 16-class mapping is not explicitly stated.

CT-SSG. CT-SSG is a supervised 2.5D ResNet-18 and spectral-graph classifier trained in five folds on CT-RATE [17]. Evaluation uses the 16-label harmonization described above. The source reports 1,334 RAD-ChestCT patients but not the exact number of evaluated volumes; all values are five-fold means.

Pillar-0. Pillar-0’s Atlas chest encoder was contrastively pretrained on 86,411 private chest CT scans [14]. The later DCP-PD study freezes Atlas, pools its multiscale representation, and trains the DCP-1 linear multilabel classifier on CT-RATE [24]. DCP-1 is evaluated on all 3,630 RAD-ChestCT volumes using this 16-label harmonization.

5.3.2. RAD-ChestCT report generation

The public RAD-ChestCT release does not include paired reference report text. Consequently, generated reports cannot be evaluated with reference-dependent text or semantic metrics such as BLEU, GREEN, RadGraph F1, or RadFact-CT F1. They can nevertheless be evaluated for clinical content by extracting the harmonized abnormalities from each generated report and comparing them with the released binary labels. Table 8 separates this report-derived clinical evaluation from direct image classification.

Table 8 | Published RAD-ChestCT report-derived evaluation. Generated reports are converted to abnormality predictions and compared with the public binary labels. Macro F1, precision, and recall are unweighted means of the corresponding positive-class per-label metrics. Because reference reports are unavailable, BLEU, ROUGE, GREEN, and other reference-dependent report metrics cannot be computed.

Method	Evaluation details	Macro F1 ↑	Precision ↑	Recall ↑
CT-CHAT [1]		0.182	0.382	0.171
BTB3D [21]	BTB3D-16 variant.	0.266	0.272	0.329
RadFM [29]	CT-CHAT [1] evaluation using the official pretrained checkpoint.	0.069	0.283	0.044
M3D-LaMed [30, 25]	EXACT evaluation [25] using official weights; modified 16 labels harmonization.	0.113	0.269	0.080
DCP-PD [24]	DCP-PD base VLM.	0.395	0.443	0.418
EXACT-CHAT [25]	modified 16 label harmonization	0.289	0.469	0.298
NV-REASON-CT (ours)	Default structured-report prompt; no RAD-ChestCT tuning. Findings and Impression were classified with the frozen official CT-RATE RadBERT at 0.5; 288 reports exceeded its 512-token limit and were truncated.	0.4854	0.5101	0.4943
	Default structured-report prompt, but the labels were parsed from the answer block directly (without RadBERT)	0.5105	0.5158	0.5524

NV-REASON-CT. We generate reports for all 3,630 public volumes with the default structured-report prompt and without RAD-ChestCT-specific tuning. A frozen CT-RATE RadBERT applied to Findings and Impression yields macro F1 0.4854, precision 0.5101, and recall 0.4943; 288 reports exceed its 512-token limit and are truncated. Parsing the final abnormality list directly from the same generated reports yields macro F1 0.5105, precision 0.5158, and recall 0.5524. See Table 8.

CT-CHAT. CT-CHAT combines the CT-CLIP volumetric encoder with Llama 3.1 through a learned multimodal projector; the projector is first aligned to CT-RATE report text and is subsequently optimized together with low-rank language-model adaptation on CT-RATE instruction data [1]. The reported RAD-ChestCT experiment uses the Llama 3.1-70B configuration on all 3,630 public volumes. Abnormalities extracted from the generated reports are compared with the 16 harmonized binary targets, yielding macro F1 0.182, precision 0.382, and recall 0.171.

BTB3D. BTB3D learns discrete volumetric tokens with a causal 3D convolutional encoder–decoder, Haar-wavelet compression, and lookup-free quantization, and projects these tokens to a Llama 3.1-8B report decoder [21]. We report its best $16 \times 16 \times 8$ tokenization variant under the CT-CHAT-16 evaluation protocol. The paper presents RAD-ChestCT as the external test set but does not state the exact evaluated number of volumes; it reports macro F1 0.266, precision 0.272, and recall 0.329.

RadFM. RadFM is a generalist medical VLM comprising a 3D ViT, a Perceiver-style token resampler, and a MedLLaMA-13B language model [29]. The RAD-ChestCT result is not a native RadFM evaluation: the

CT-CHAT study applies the official pretrained checkpoint without retraining to all 3,630 public volumes and evaluates its generated reports with the CT-CHAT-16 label protocol [1]. This later evaluation gives macro F1 0.069, precision 0.283, and recall 0.044.

M3D-LaMed. M3D-LaMed pairs a native 3D ViT with a Llama-family language model and is trained by image-text contrastive alignment followed by multimodal instruction tuning on M3D-Data [30]. Its original publication does not evaluate RAD-ChestCT. The values in Table 8 instead come from the later EXACT study, which runs the official weights on all 3,630 volumes and uses its modified 16-label harmonization and RadBERT report labeling [25]. It reports macro F1 0.113 (95% CI, 0.091–0.137), precision 0.269, and recall 0.080.

DCP-PD base VLM. DCP-PD builds its report generator from the frozen Pillar-0 Atlas visual backbone, an attention-pooling projector, and Llama 3-Instruct-8B, and trains the projector and language-model adapters on structured CT-RATE reports [24]. On all 3,630 RAD-ChestCT volumes under the 16-label protocol, this base model achieves macro F1 0.395, precision 0.443, and recall 0.418.

EXACT-CHAT. EXACT learns 18 voxel-level anomaly-aware maps with a Y-Mamba image backbone; EXACT-CHAT connects the frozen encoder to Llama 3.1-8B-Instruct through an attentional pooling projector and also supplies its map-derived diagnostic predictions as text [25]. It is trained on CT-RATE and evaluated without RAD-ChestCT adaptation on all 3,630 external volumes. Under the modified 16-label protocol, the unrefined model listed here achieves RadBERT macro F1 0.289, precision 0.469, and recall 0.298.

5.4. Internal Dataset Evaluation

We additionally evaluate NV-REASON-CT on a held-out internal NIH collection. The dataset contains 5,092 cases from a unique patients. Evaluation is coverage-aware: 3,981 cases contain the full chest and 4,205 contain the full abdomen, with 3,132 cases contributing to both regional cohorts. Reports were labeled for 30 chest and 29 abdominal findings. We balance positive vs negative findings via subsampling. This produced 35,992 balanced chest case-finding examples and 37,740 balanced abdomen case-finding examples. The class-specific cohorts vary substantially in size: for example, the balanced evaluation includes only 22 examples for appendiceal abnormality, 56 for pneumothorax, 58 for pneumoperitoneum, and 86 for pulmonary embolism. Macro-F1 is the unweighted mean of the positive-class F1 scores over the findings in each region. Table 9 reports the per-abnormality and macro results, while Figure 8 visualizes the corresponding balanced F1 profiles by anatomical system. For AUROC, thinking was disabled and the normalized next-token probability of *Yes* was retained as a continuous score. The same balanced selections were used for both metrics; no threshold tuning was performed.

Table 9 | Per-abnormality NV-REASON-CT classification results on the internal NIH validation cohort. Positive and negative cases are balanced independently for each finding. Macro denotes the unweighted mean over findings in each anatomical cohort. Pos. gives the number of positive cases retained after balancing.

Chest cohort (30 findings)				Abdomen cohort (29 findings)			
Finding	Pos.	F1 ↑	AUROC ↑	Finding	Pos.	F1 ↑	AUROC ↑
Macro	–	0.6302	0.8113	Macro	–	0.5759	0.7615
Medical material	1847	0.8405	0.8831	Liver cyst	398	0.6420	0.7657
Arterial wall calcification	911	0.6984	0.8353	Liver mass or non-cystic lesion	1853	0.7271	0.7963
Cardiomegaly	339	0.4848	0.7586	Hepatic steatosis	342	0.6024	0.8058
Pericardial effusion	343	0.6238	0.8099	Biliary dilation	367	0.5219	0.7644
Coronary artery wall calcification	827	0.7443	0.8587	Gallstone or gallbladder abnormality	519	0.5831	0.7578
Hiatal hernia	243	0.5666	0.7819	Splenomegaly or splenic lesion	755	0.6933	0.7976
Lymphadenopathy	1735	0.7114	0.7690	Pancreatic lesion or duct abnormality	454	0.6371	0.7570
Emphysema	169	0.7483	0.8643	Pancreatitis	52	0.3824	0.8023
Atelectasis	1202	0.6804	0.7758	Renal cyst	1134	0.6115	0.7474
Lung nodule	1648	0.6614	0.7136	Renal mass	743	0.5214	0.6590
Lung opacity	655	0.6442	0.7978	Urinary calculus	384	0.4944	0.7111
Pulmonary fibrotic sequela	1096	0.6831	0.7872	Urinary obstruction	211	0.7416	0.8716
Pleural effusion	481	0.9207	0.9656	Adrenal nodule or mass	940	0.5070	0.6587
Mosaic attenuation pattern	89	0.5000	0.8514	Bowel wall thickening or mass	534	0.5012	0.6433
Peribronchial thickening	178	0.3571	0.7656	Bowel obstruction or dilation	107	0.4552	0.8080
Consolidation	257	0.7586	0.9040	Diverticular disease	554	0.5207	0.6993
Bronchiectasis	383	0.5046	0.7473	Appendiceal abnormality	11	0.1667	0.6322
Interlobular septal thickening	107	0.4058	0.8207	Ascites	670	0.7328	0.8637
Lung mass	333	0.7923	0.8867	Peritoneal/omental disease	467	0.6631	0.8279
Pleural thickening or mass	544	0.6411	0.7708	Pneumoperitoneum	29	0.5714	0.8966
Mediastinal mass	188	0.7853	0.8891	Abdominal vascular abnormality	510	0.5732	0.7266
Lymphoma	458	0.3351	0.7190	Abdominal vascular calcification	1272	0.7011	0.8193
Pneumothorax	28	0.6977	0.9700	Abdominal lymphadenopathy	1365	0.7057	0.7823
Thoracic bone lesion	531	0.6339	0.7399	Abdominal wall hernia	379	0.3044	0.6411
Thoracic fracture	208	0.5413	0.8005	Abdominal soft tissue mass	941	0.6590	0.7552
Chest wall or breast mass	404	0.5752	0.7673	Abdominopelvic bone lesion	892	0.6098	0.7032
Pulmonary embolism	43	0.2800	0.6723	Abdominopelvic fracture	190	0.3770	0.7305
Thoracic vascular abnormality	742	0.6499	0.7614	Abdominal postoperative change	1913	0.7709	0.8361
Esophageal abnormality	387	0.5524	0.7471	Abdominal medical material	884	0.7241	0.8233
Postoperative or treatment-related change	1620	0.8874	0.9250				

Performance on the balanced NIH cohorts varied substantially across findings. In the chest, the highest F1 scores were observed for pleural effusion, postoperative or treatment-related change, and medical material. In the abdomen, postoperative change, urinary obstruction, ascites, and liver masses or non-cystic lesions yielded the highest F1 scores. Lower F1 scores were observed for pulmonary embolism and lymphoma in the chest, and for appendiceal abnormality and abdominal wall hernia in the abdomen. These findings highlight variation within each anatomical region; estimates for findings with small evaluation cohorts should be interpreted cautiously.

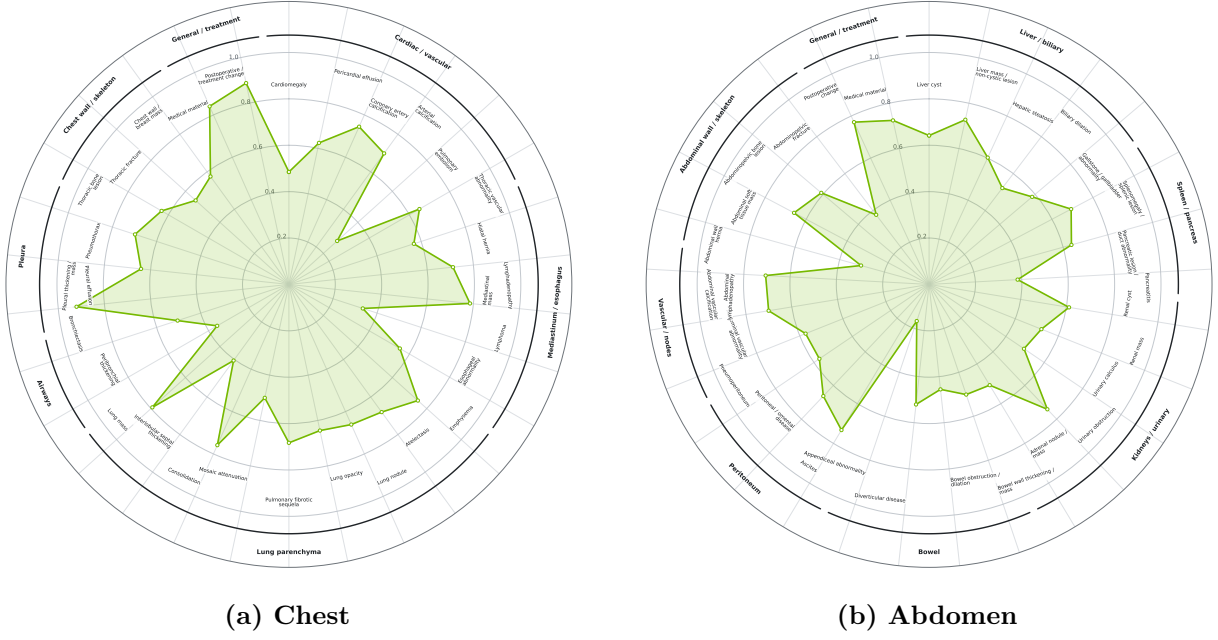


Figure 8 | Per-abnormality positive-class F1 on the balanced NIH validation set cohorts: (a) 30 chest findings and (b) 29 abdominal findings. Findings are grouped by anatomical system, and each spoke reports the F1 value from the independently balanced positive-negative cohort used in Table 9.

5.5. Subjective Evaluation with Expert Radiologists

We conducted a preliminary within-subject reader study with US board-certified radiologists to assess perceived output quality, usefulness, time savings, and influence on clinical judgment. For each case, readers first interpreted the CT without AI assistance, following their usual clinical workflow. After a short distraction, they reviewed the same case with the model’s full reasoning text and structured report. Readers could copy the structured report, verify its contents, and edit it as needed.

Participants and Materials.

- **Participants:** 2 radiologists with 10+ years of clinical practice.
- **Cases:** 10 cases comprising 2 normal and 8 abnormal examinations, with 5 chest and 5 abdominal CT cases.
- **Platform:** Web interface of the trained NV-REASON-CT model.

Questionnaire Items The ten agreement items were rated using a 5-point **Likert scale**: 1 = *Strongly disagree*, 2 = *Disagree*, 3 = *Neither agree nor disagree*, 4 = *Agree*, 5 = *Strongly agree*.

A. Accuracy and Reasoning Quality

- The reasoning trace includes correct statements about key observations.
- The reasoning does not include speculative or unsupported statements.
- The AI identified a relevant finding I initially missed.
- Uncertainty and limitations are explicitly stated when appropriate.

B. Time & Efficiency

- The AI output reduced the time required to compose my report.

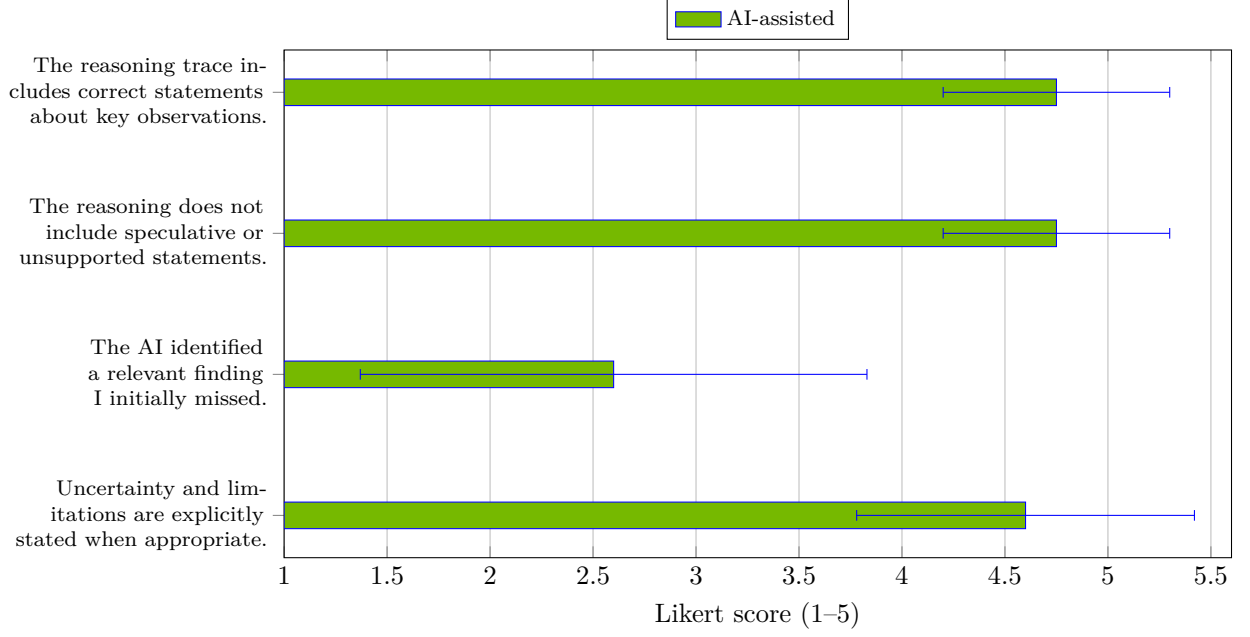


Figure 9 | The results (mean/std) of Accuracy and Reasoning Quality survey. Expert radiologists were tasked to write a report given the AI output. Likert scale: 1 - Strongly Disagree, 2 - Disagree, 3 - Neither Agree Nor Disagree, 4 - Agree, 5 - Strongly Agree. Ratings were high for correctness, avoidance of unsupported statements, and acknowledgment of uncertainty, and lower for identifying initially missed findings.

- The AI output helped me reach a decision faster.
- Approximate percentage of time saved on this case (percentage calculated)

C. Trust and Confidence

- After reading the AI output, I feel more confident (less uncertain) in my decision.
- If the AI had disagreed with me on this case, I would re-evaluate and re-examine the study.
- The AI's suggestions improved my final decision.
- The AI did not bias me toward incorrect conclusions.

5.5.1. Reader-assessment results

Accuracy and Reasoning Quality. Accuracy and Reasoning Quality ratings are shown in Figure 9.

- **The reasoning trace includes correct statements about key observations.** Ratings were predominantly *Agree* or *Strongly Agree* (4.75 ± 0.55).
- **The reasoning does not include speculative or unsupported statements.** Ratings were predominantly *Agree* or *Strongly Agree* (4.75 ± 0.55).
- **The AI identified a relevant finding I initially missed.** Ratings were lower for this item (2.60 ± 1.23). The radiologists attributed this to their careful initial review of each case, reporting that they did not miss any findings during their manual review. Whether this differs under routine clinical workloads of many daily cases remains to be evaluated.
- **Uncertainty and limitations are explicitly stated when appropriate.** Most responses were *Agree* or *Strongly Agree* (4.60 ± 0.82).

Time & Efficiency. Figure 10 compares the average reported time without and with AI assistance. Readers could copy and edit the AI-generated structured reports as needed. The mean time was 26.25 min without AI and 13.13 min with AI, corresponding to a 50% reduction.

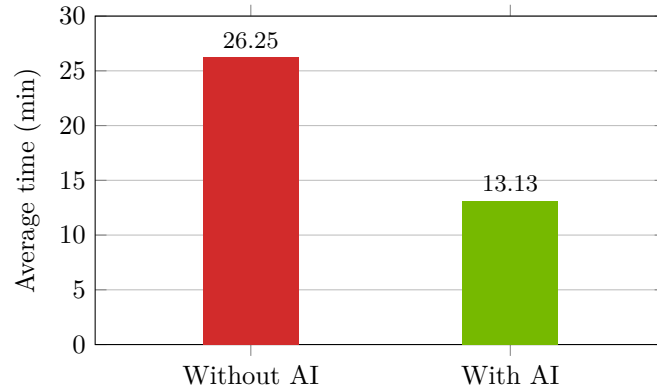


Figure 10 | Average reported time spent interpreting a CT and preparing a report without and with AI assistance.

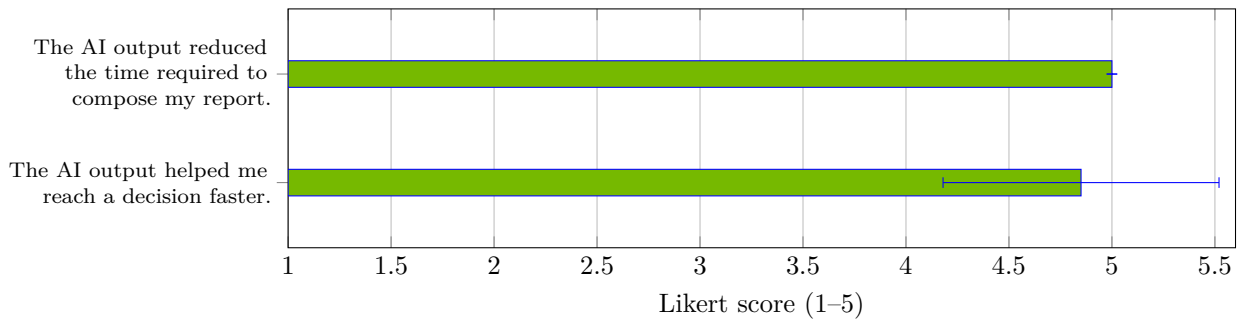


Figure 11 | The Time & Efficiency survey (mean/std). Likert scale: 1 - Strongly Disagree, 2 - Disagree, 3 - Neither Agree Nor Disagree, 4 - Agree, 5 - Strongly Agree.

The Time & Efficiency survey ratings are shown in Figure 11.

- **The AI output reduced the time required to compose my report.** All responses were *Strongly Agree* (5.00 ± 0.00).
- **The AI output helped me reach a decision faster.** Nineteen responses were *Strongly Agree* and one was *Disagree* (4.85 ± 0.67).

Trust & Confidence. The Trust & Confidence survey ratings are shown in Figure 12.

- **After reading the AI output, I feel more confident (less uncertain) in my decision.** Ratings were predominantly *Agree* or *Strongly Agree* (4.70 ± 0.57).
- **If the AI had disagreed with me on this case, I would re-evaluate and re-examine the study.** All responses were *Strongly Agree* (5.00 ± 0.00).
- **The AI's suggestions improved my final decision.** All responses were *Agree* or *Strongly Agree* (4.70 ± 0.47).
- **The AI did not bias me toward incorrect conclusions.** Most responses were *Strongly Agree* (4.95 ± 0.22).

Overall, radiologists reported favorable assessments of output quality, report-writing efficiency, and confidence in their decisions. These findings represent preliminary reader perceptions; diagnostic accuracy and clinical safety require separate evaluation.

6. Limitations

We acknowledge the following limitations:

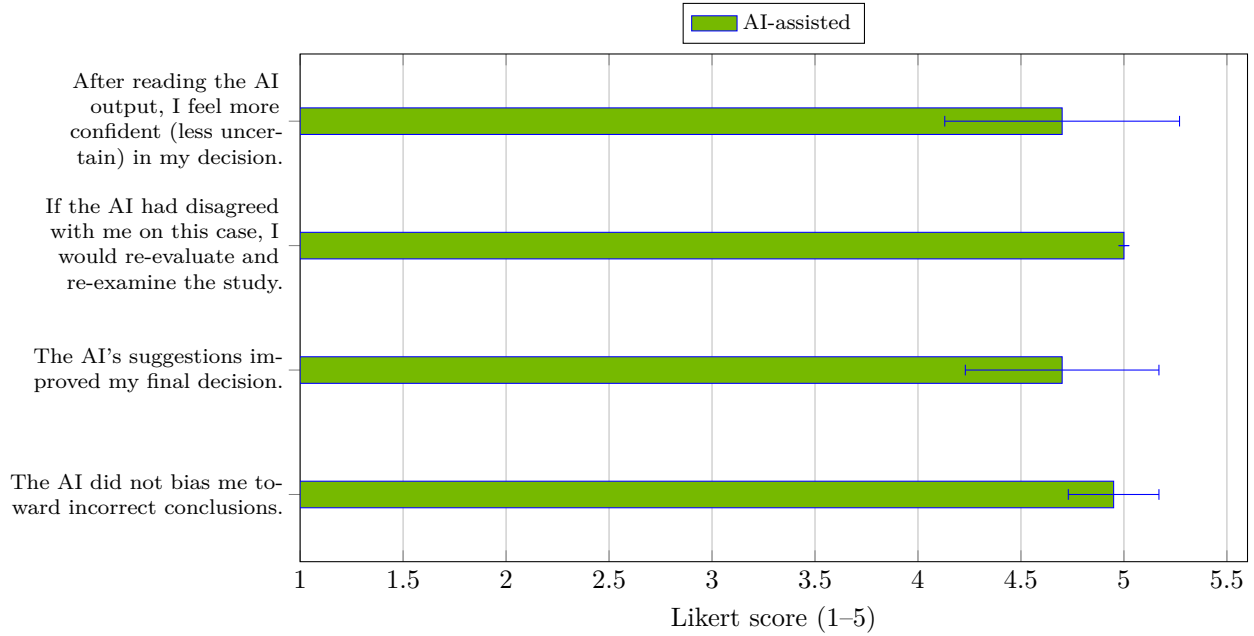


Figure 12 | The Trust & Confidence survey (mean/std). Likert scale: 1 - Strongly Disagree, 2 - Disagree, 3 - Neither Agree Nor Disagree, 4 - Agree, 5 - Strongly Agree.

- **Hallucinations and reasoning reliability.** As with other generative LLM and VLM systems, NV-REASON-CT can produce findings or explanations that are not supported by the image. Generated reasoning provides a reviewable explanation, but does not establish that the text faithfully represents the model’s internal computation.
- **Anatomical scope and cropping.** The model is designed for chest and abdominal CT and processes one fixed-size regional crop at a time. The automatic cropping heuristic relies on detecting a sufficient portion of the lungs and can select a suboptimal field of view in scans with limited or unusual coverage. Findings outside the selected crop cannot be recovered by the language model. More robust segmentation, learned localization, or multi-region processing may be needed for whole-body examinations and other anatomical regions.
- **Data and reasoning supervision.** Although training combines several CT sources and task types, it does not capture the full diversity of scanners, protocols, institutions, patient populations, and disease presentations encountered in clinical practice. Rare findings remain underrepresented, and report-derived labels and targets inherit ambiguity from their source reports. Direct expert reasoning supervision comes from a limited set of recorded and transcribed CT interpretations; the expanded corpus includes synthetic narrated interpretations derived from source reports using these expert examples. Larger multi-institutional datasets with more directly authored and reviewed expert reasoning may improve robustness.
- **Clinical context and prior examinations.** The reported evaluations primarily condition the model on the CT volume and task instruction. Clinical indications, laboratory results, patient history, and prior examinations are not incorporated systematically. Radiologists use this information to resolve uncertainty and assess interval change, and its principled integration is an important direction for future work.
- **Evaluation and clinical validation.** Automated results depend on dataset-specific label ontologies, report formats, and evaluator definitions, as well as differences in evaluation cohorts, prompting, and model adaptation, limiting direct comparison across studies. The reported evaluations assess the complete model; controlled ablations are needed to isolate the contributions of the spatial interface, reasoning supervision, and GRPO. The reader study is preliminary and reflects reader perceptions. Because unaided review always preceded AI-assisted review of the same case, recall and reading-order effects may influence the reported benefits. It is not a prospective assessment of clinical safety or effectiveness. Larger, multi-center studies are required to evaluate calibration, failure modes, workflow

impact, and generalization in routine practice.

7. Conclusion

We introduced NV-REASON-CT, a generative vision-language model for native 3D interpretation of chest and abdominal CT. The model uses a native 3D vision encoder and passes all visual tokens, together with their explicit 3D positions, to the language model for joint processing with text. This architecture preserves volumetric spatial context throughout the vision-language interface and supports abnormality classification, report generation, and interactive questions with reviewable clinical explanations.

A central contribution of this work is the curated, diverse multimodal training corpus spanning chest and abdominal CT. It combines structured reports, abnormality-focused and anatomy-specific questions, multi-turn interactions, and refusal examples. A limited set of recorded and transcribed expert CT interpretations provides direct supervision and serves as exemplars for generating additional report-grounded synthetic narrations. These annotations are designed to teach the model to communicate supporting observations, diagnostic considerations, and uncertainty. Training follows two stages: end-to-end supervised fine-tuning on the complete task mixture, followed by GRPO using region-aware, verifiable rewards for finding agreement and report structure.

We evaluated NV-REASON-CT across CT-RATE, Merlin, RAD-ChestCT, and a held-out internal NIH cohort, covering classification and report generation across different datasets, anatomical regions, and label ontologies. On CT-RATE, the model achieved strong classification and report-generation performance without a task-specific classification head. In a preliminary study, expert radiologists reported increased confidence with AI-assisted review, alongside a 50% reduction in average reported interpretation and reporting time. By releasing NV-REASON-CT together with its SFT and GRPO training code, we aim to support further research and independent evaluation of expert-guided 3D CT interpretation.

A. Per-abnormality CT-RATE classification results

Tables A1 and A2 expand the macro-level comparison in Table 3 by reporting F1 and AUROC, respectively, for each of the 18 CT-RATE abnormalities. Each table is divided into two parts to retain the complete comparator set at a readable scale. They are included to make label-specific performance and differences in checkpoints, evaluation units, and adaptation protocols auditable without interrupting the main Results narrative.

Table A1 | Per-abnormality positive-class F1 on the CT-RATE public validation set (Part 1 of 2).

Abnormality	Proposed	CT-CLIP			Merlin		COLIPRI
	NV-Reason-CT (ours)	Ours (Zero-shot)	Ours (VocabFine)	VoxelFM eval.	Ours (official ckpt)	VoxelFM eval.	Ours (released ckpt)
Medical material	0.7270	0.2672	0.0364	0.1763	0.1956	0.5607	0.2677
Arterial wall calcification	0.7750	0.6546	0.6475	0.4452	0.5436	0.7278	0.7337
Cardiomegaly	0.6390	0.3330	0.5303	0.2300	0.2733	0.4817	0.5689
Pericardial effusion	0.5780	0.2136	0.0000	0.1532	0.2202	0.3028	0.4263
Coronary artery wall calcification	0.7770	0.6139	0.6029	0.4419	0.5294	0.6890	0.6550
Hiatal hernia	0.5410	0.3369	0.0828	0.2444	0.2644	0.3574	0.3565
Lymphadenopathy	0.5750	0.5082	0.3003	0.3728	0.3881	0.4734	0.4181
Emphysema	0.5270	0.4435	0.2828	0.3413	0.3395	0.4676	0.3706
Atelectasis	0.6040	0.4362	0.3473	0.3482	0.3012	0.4880	0.5069
Lung nodule	0.6590	0.5229	0.5781	0.4738	0.5845	0.6181	0.6332
Lung opacity	0.7450	0.5504	0.3262	0.4501	0.5668	0.6168	0.6796
Pulmonary fibrotic sequela	0.5410	0.4104	0.2130	0.0000	0.3420	0.4839	0.4954
Pleural effusion	0.8590	0.3863	0.6110	0.2644	0.5674	0.7135	0.8302
Mosaic attenuation pattern	0.4520	0.2545	0.0472	0.1653	0.1554	0.3252	0.2176
Peribronchial thickening	0.4410	0.2977	0.0000	0.2262	0.2383	0.3766	0.2591
Consolidation	0.7120	0.4398	0.1696	0.2835	0.4551	0.5395	0.5807
Bronchiectasis	0.4680	0.2397	0.0000	0.1879	0.1850	0.2929	0.2226
Interlobular septal thickening	0.4290	0.2476	0.0160	0.1627	0.2920	0.4043	0.2679
Macro-F1	0.6140	0.3976	0.2662	0.2760	0.3579	0.4955	0.4717

Table A1 | Per-abnormality positive-class F1 on the CT-RATE public validation set (Part 2 of 2).

Abnormality	VoxelFM	CT-SSG	RadFM	M3D-LaMed	Jolia	Pillar-0 (DCP-1)	MedGemma 1.5	ClinFusion 8B	ClinFusion 32B
Medical material	0.6269	0.6310	0.3402	0.3579	0.2525	0.5940	0.2605	0.3731	0.1829
Arterial wall calcification	0.7849	0.7780	0.6474	0.6441	0.5248	0.7430	0.5392	0.6579	0.6386
Cardiomegaly	0.5410	0.6240	0.5279	0.5216	0.2222	0.5640	0.4415	0.4441	0.3909
Pericardial effusion	0.4916	0.5540	0.2557	0.3090	0.2228	0.3800	0.0090	0.3189	0.1172
Coronary artery wall calcification	0.7663	0.7670	0.6373	0.5030	0.4817	0.7180	0.4344	0.5827	0.5747
Hiatal hernia	0.4125	0.4210	0.3197	0.3147	0.2733	0.3860	0.2439	0.2439	0.3294
Lymphadenopathy	0.5554	0.5200	0.4607	0.4330	0.4112	0.3680	0.3976	0.4825	0.0743
Emphysema	0.5356	0.5090	0.4322	0.3559	0.3595	0.5040	0.3350	0.4007	0.3668
Atelectasis	0.5537	0.5210	0.4096	0.4123	0.4046	0.6060	0.3802	0.4491	0.3579
Lung nodule	0.6661	0.6370	0.5915	0.6180	0.4802	0.6960	0.1659	0.6348	0.6214
Lung opacity	0.7194	0.7380	0.5873	0.5723	0.6193	0.7910	0.6037	0.6252	0.6345
Pulmonary fibrotic sequela	0.5405	0.4670	0.3183	0.4407	0.4208	0.3510	0.1103	0.4349	0.4437
Pleural effusion	0.8400	0.8250	0.6158	0.6257	0.3823	0.8470	0.4884	0.5195	0.4100
Mosaic attenuation pattern	0.5090	0.4560	0.3114	0.2970	0.2509	0.3980	0.1631	0.3174	0.0394
Peribronchial thickening	0.4017	0.3940	0.2755	0.2850	0.2292	0.4020	0.2078	0.3398	0.3556
Consolidation	0.6700	0.6410	0.4636	0.4387	0.5275	0.6690	0.2468	0.5009	0.5088
Bronchiectasis	0.4013	0.3910	0.2492	0.2603	0.2044	0.3830	0.2028	0.2740	0.3017
Interlobular septal thickening	0.4374	0.4180	0.3409	0.2732	0.1925	0.3860	0.2285	0.3551	0.3870
Macro-F1	0.5807	0.5718	0.4325	0.4257	0.3589	0.5440	0.3033	0.4419	0.3741

Table A2 | Per-abnormality AUROC on the CT-RATE public validation set (Part 1 of 2).

Abnormality	Proposed	CT-CLIP			Merlin		COLIPRI
	NV-Reason-CT (ours)	Ours (Zero-shot)	Ours (VocabFine)	VoxelFM eval.	Ours (official ckpt)	VoxelFM eval.	Ours (released ckpt)
Medical material	0.9520	0.7172	0.7532	0.5453	0.6136	0.8815	0.7357
Arterial wall calcification	0.9280	0.8557	0.8758	0.6973	0.7458	0.9016	0.9317
Cardiomegaly	0.9420	0.8582	0.9200	0.6037	0.7901	0.9172	0.9297
Pericardial effusion	0.9200	0.7728	0.7818	0.5620	0.7526	0.8206	0.8479
Coronary artery wall calcification	0.9350	0.8533	0.8600	0.6786	0.7433	0.8976	0.9220
Hiatal hernia	0.8430	0.7055	0.7208	0.5722	0.5713	0.7318	0.7108
Lymphadenopathy	0.7810	0.7001	0.7058	0.5569	0.6013	0.6993	0.7552
Emphysema	0.8130	0.7426	0.7518	0.5835	0.6437	0.7545	0.7690
Atelectasis	0.8240	0.6727	0.6904	0.5812	0.5949	0.7342	0.7744
Lung nodule	0.7800	0.5716	0.6525	0.5144	0.4944	0.6487	0.7065
Lung opacity	0.8730	0.6401	0.6948	0.5547	0.6720	0.7451	0.8056
Pulmonary fibrotic sequela	0.7650	0.5778	0.6479	0.5071	0.5389	0.6732	0.7055
Pleural effusion	0.9750	0.9074	0.9344	0.6146	0.9136	0.9370	0.9668
Mosaic attenuation pattern	0.8960	0.7745	0.7781	0.5476	0.5913	0.8331	0.7777
Peribronchial thickening	0.8130	0.6913	0.7094	0.5898	0.6150	0.7829	0.8210
Consolidation	0.9280	0.7307	0.7508	0.5253	0.7241	0.8133	0.9007
Bronchiectasis	0.8180	0.6536	0.6138	0.5378	0.5451	0.7436	0.7976
Interlobular septal thickening	0.8870	0.7655	0.8077	0.5593	0.7593	0.8550	0.8319
Macro-AUROC	0.8710	0.7328	0.7583	0.5740	0.6617	0.7983	0.8161

Table A2 | Per-abnormality AUROC on the CT-RATE public validation set (Part 2 of 2).

Abnormality	VoxelFM	CT-SSG	RadFM	M3D-LaMed	Jolia	Pillar-0 (DCP-1)
Medical material	0.9423	0.9050	0.7402	0.7749	0.7319	0.9340
Arterial wall calcification	0.9436	0.9280	0.8730	0.8351	0.7444	0.9140
Cardiomegaly	0.9358	0.9350	0.9096	0.9009	0.8011	0.9370
Pericardial effusion	0.8963	0.9020	0.7953	0.7987	0.7939	0.8580
Coronary artery wall calcification	0.9420	0.9310	0.8562	0.8202	0.7504	0.9300
Hiatal hernia	0.8140	0.7730	0.7313	0.7071	0.6032	0.7590
Lymphadenopathy	0.7700	0.7430	0.6960	0.6659	0.5843	0.8830
Emphysema	0.8367	0.7890	0.7310	0.6971	0.6896	0.8110
Atelectasis	0.8057	0.7760	0.6837	0.6763	0.6404	0.8050
Lung nodule	0.7863	0.6260	0.6322	0.5936	0.5096	0.7940
Lung opacity	0.8849	0.8660	0.7079	0.6335	0.7302	0.8850
Pulmonary fibrotic sequela	0.7500	0.6830	0.6227	0.6188	0.5497	0.7060
Pleural effusion	0.9738	0.9690	0.9323	0.9115	0.8991	0.9820
Mosaic attenuation pattern	0.9057	0.8840	0.8009	0.8190	0.7849	0.8730
Peribronchial thickening	0.8043	0.7980	0.7218	0.6959	0.5865	0.8140
Consolidation	0.9210	0.9010	0.7410	0.7267	0.8228	0.9260
Bronchiectasis	0.8495	0.7870	0.6786	0.6832	0.5604	0.7890
Interlobular septal thickening	0.9007	0.8600	0.8113	0.7673	0.7914	0.8920
Macro-AUROC	0.8701	0.8364	0.7592	0.7403	0.6986	0.8610

B. Per-abnormality CT-RATE report-derived results

Table B1 reports the positive-class F1 of each CT-RATE abnormality after labels are extracted from generated reports. These values quantify finding coverage in report text and are distinct from the direct image-classification results in Tables A1 and A2. The CT2Rep, CT-CHAT, and BTB3D-16 columns reproduce BTB3D Table 5 [21]; MS-VLM reproduces Table E.1 of that study [22]; and MonteRET reproduces Supplementary Table S4 [26]. COLIPRI-CRM and VoxelFM do not provide numerical per-abnormality tables, so their entries were digitized from the authors’ vector and raster figures, respectively, and should be regarded as approximate [3, 15]. Because these studies use non-identical cohorts and report-extraction implementations, the columns document available evidence rather than a controlled ranking.

Table B1 | Per-abnormality positive-class F1 derived from generated reports on CT-RATE. Columns marked † were digitized from author-provided figures and are approximate; the corresponding Macro-F1 values are the authors’ reported aggregate results.

Abnormality	CT2Rep	CT-CHAT	BTB3D 16×16×8	MS-VLM	COLIPRI CRM [†]	VoxelFM [†]	MonteRET	NV-REASON-CT (ours)
Medical material	0.000	0.006	0.142	0.157	0.414	0.234	0.103	0.679
Arterial wall calcification	0.322	0.451	0.414	0.334	0.682	0.636	0.700	0.766
Cardiomegaly	0.013	0.123	0.305	0.271	0.481	0.488	0.508	0.585
Pericardial effusion	0.000	0.009	0.095	0.162	0.223	0.244	0.380	0.561
Coronary artery wall calcification	0.335	0.412	0.403	0.321	0.642	0.588	0.630	0.756
Hiatal hernia	0.074	0.207	0.164	0.179	0.199	0.282	0.480	0.531
Lymphadenopathy	0.013	0.069	0.358	0.319	0.451	0.311	0.198	0.545
Emphysema	0.198	0.391	0.196	0.282	0.422	0.464	0.420	0.521
Atelectasis	0.323	0.341	0.269	0.279	0.489	0.481	0.365	0.597
Lung nodule	0.029	0.443	0.427	0.499	0.538	0.605	0.536	0.687
Lung opacity	0.557	0.266	0.408	0.505	0.680	0.665	0.331	0.734
Pulmonary fibrotic sequela	0.104	0.069	0.318	0.327	0.453	0.286	0.404	0.491
Pleural effusion	0.341	0.173	0.308	0.187	0.724	0.676	0.272	0.847
Mosaic attenuation pattern	0.198	0.064	0.183	0.164	0.296	0.425	0.278	0.457
Peribronchial thickening	0.099	0.000	0.125	0.194	0.267	0.273	0.239	0.370
Consolidation	0.236	0.120	0.259	0.287	0.521	0.569	0.278	0.682
Bronchiectasis	0.013	0.091	0.126	0.111	0.334	0.208	0.260	0.461
Interlobular septal thickening	0.032	0.075	0.135	0.125	0.258	0.353	0.190	0.392
Macro-F1	0.160	0.184	0.258	0.261	0.449	0.432	0.365	0.592

C. Per-abnormality Merlin classification results

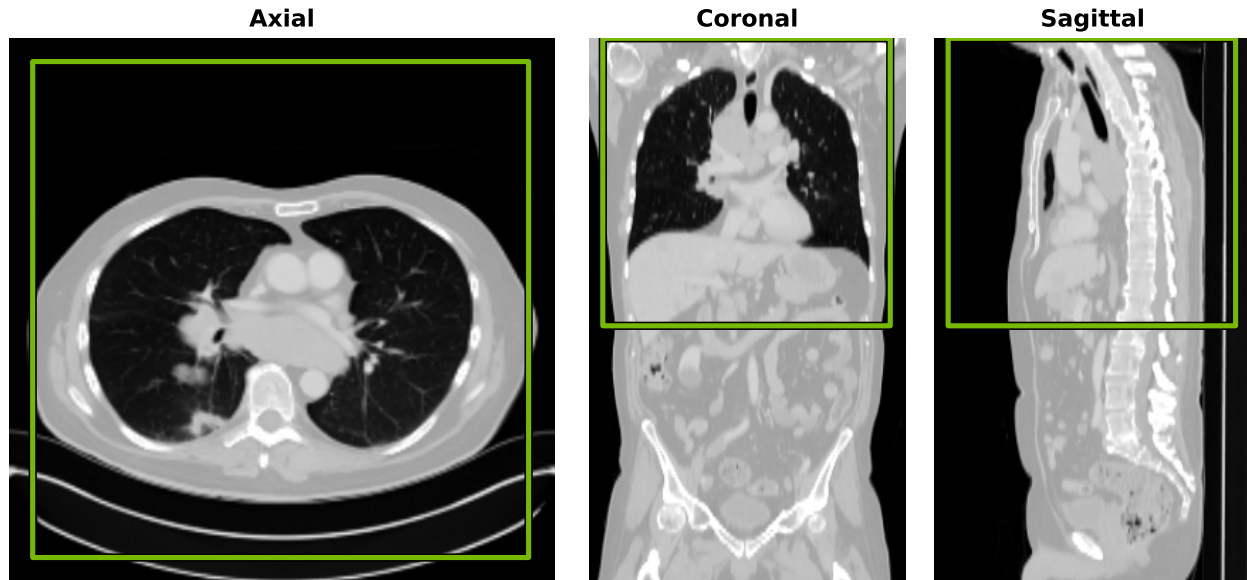
Table C1 expands the NV-REASON-CT summary in Table 5 by reporting the positive-class F1 and AUROC for each of the 30 Merlin findings. Results use the same balanced cohort and uniform prompt as the main-text result. F1 is computed at the fixed 0.5 next-token probability threshold, and AUROC is computed from the continuous normalized *Yes* probabilities. The six lower-thoracic findings specified in Section 5.2.1 use the chest anatomical region; the other 24 use the abdomen region. Prompt selection and this region policy used the released test cohort, as noted in Section 5.2.1.

Table C1 | Per-abnormality positive-class F1 and AUROC for NV-REASON-CT on the balanced Merlin test cohort. F1 uses a fixed 0.5 threshold on the normalized next-token probability of *Yes*; AUROC uses the continuous probability. The macro average is the unweighted mean across the 30 findings.

Finding	F1 at 0.5 ↑	AUROC ↑
Submucosal edema	0.7092	0.7608
Renal hypodensities	0.6901	0.7307
Aortic valve calcification	0.9078	0.9523
Coronary calcification	0.8000	0.8447
Thrombosis	0.6750	0.6743
Metastatic disease	0.7931	0.8705
Pancreatic atrophy	0.7258	0.7641
Renal cyst	0.6779	0.7877
Osteopenia	0.8102	0.9164
Surgically absent gallbladder	0.7423	0.8363
Atelectasis	0.7698	0.8349
Abdominal aortic aneurysm	0.9697	0.9931
Anasarca	0.8702	0.9193
Hiatal hernia	0.7917	0.8582
Lymphadenopathy	0.6435	0.6591
Prostatomegaly	0.7273	0.6999
Biliary ductal dilation	0.7010	0.8060
Cardiomegaly	0.8118	0.8871
Splenomegaly	0.8140	0.8982
Hepatomegaly	0.7347	0.8546
Atherosclerosis	0.8197	0.8468
Ascites	0.8859	0.9520
Pleural effusion	0.9236	0.9642
Hepatic steatosis	0.6800	0.7407
Appendicitis	0.5484	0.7331
Gallstones	0.7234	0.8220
Hydronephrosis	0.7292	0.8316
Bowel obstruction	0.7794	0.9006
Free air	0.8132	0.8741
Fracture	0.6978	0.7749
Macro average	0.7655	0.8329

D. Model Output Examples

This section presents representative outputs from NV-REASON-CT. The first example shows the full CT volume in axial, coronal, and sagittal cross-sections, with the square projection of the anatomy-aware $384 \times 384 \times 384$ -mm chest crop outlined in green. This cubic crop is the image input to the model. The visualization is followed by a structured chest CT report and a chest CT reasoning output generated from this input. User prompts are shown in green and NV-REASON-CT outputs are shown in gray.



D.1. Structured chest CT report

User: write a structured chest CT report

NV-REASON-CT:

<think> I will produce a structured chest CT report with a concise impression. </think>

TECHNIQUE: IV contrast CT.

FINDINGS:

Medical devices: Left-sided port catheter with the tip in the superior vena cava.

CHEST:

Lungs and airways: Right lung mass with multiple satellite nodules in the right lower lobe. Peripheral ground-glass opacities in the right lower lobe superior segment and few small focal ground-glass opacities in the right upper lobe. Patchy ground-glass opacities also noted in the left upper lobe and superior segment left lower lobe. These may relate to infection or inflammation or tumor spread. Central airways are patent.

Pleura: No pleural effusion or pneumothorax.

Mediastinum and hila: Multiple heterogeneously enhancing right supraclavicular, mediastinal, right hilar, posterior mediastinal, paravertebral, para-aortic, and retrocrural nodes, including bulky right supraclavicular adenopathy and bulky subcarinal adenopathy.

Heart and pericardium: Normal heart size. No pericardial effusion.

Great vessels: Normal caliber thoracic aorta and main pulmonary artery.

Esophagus: No esophageal abnormality.

Chest wall and breast: No chest wall or breast mass.

Bones: Enhancing paraspinal soft tissue mass with destruction of the T10 vertebral body.

IMPRESSION:

1. Right lung mass with multiple satellite nodules in the right lower lobe.
2. Multifocal ground-glass opacities in both lungs, which may relate to infection or inflammation or tumor spread.
3. Extensive right supraclavicular, mediastinal, right hilar, posterior mediastinal, paravertebral, para-aortic, and retrocrural adenopathy.
4. Enhancing paraspinal soft tissue mass with destruction of the T10 vertebral body.

User: Could the bilateral ground-glass opacities represent infection rather than tumor spread? Explain the evidence and uncertainty.

NV-REASON-CT:

<think> The question asks whether infection is a plausible explanation for the ground-glass opacities. Ground-glass opacity can reflect inflammation, infection, edema, hemorrhage, or tumor-related change depending on context. I inspect the CT image and see multifocal bilateral ground-glass opacities with a right lung mass and extensive adenopathy. The imaging pattern is not specific, so infection cannot be confirmed from CT alone. </think>

Yes, infection is a plausible possibility, especially for the peripheral ground-glass opacities. However, the CT appearance is nonspecific, and tumor-related inflammation or spread cannot be excluded given the lung mass and extensive adenopathy.

User: The patient has fever and cough. How does this new clinical information change your assessment?

NV-REASON-CT:

<think> The question asks how symptoms affect the differential. Fever and cough can be seen with infection, inflammatory tumor spread, or treatment-related change. I review the CT and see bilateral ground-glass opacities along with a right lung mass and extensive adenopathy. Imaging alone cannot determine the exact cause of the symptoms. The new clinical information makes infection a real consideration but does not exclude malignancy. </think>

It raises infection as a plausible explanation for the bilateral ground-glass opacities, especially if clinically apparent. However, the CT also shows a right lung mass and extensive adenopathy, so tumor-related spread or treatment-related change remains possible.

User: Where is the dominant lung mass?

NV-REASON-CT:

<think> The question asks for the location of the main lung mass. I review the lung windows and focus on the dominant mass-like abnormality. The CT shows the main mass centered in the right lower lobe. There are also satellite nodules in the right lower lobe and additional small focal opacities elsewhere. </think>

The dominant lung mass is in the right lower lobe, with several satellite nodules in the same lobe.

User: Which nodal regions contain lymphadenopathy?

NV-REASON-CT:

<think> The question asks for the distribution of enlarged lymph nodes. I inspect the mediastinum, hila, supraclavicular regions, and posterior mediastinum on the CT. The CT shows extensive adenopathy in multiple thoracic and lower neck stations. The most prominent nodal disease is in the right supraclavicular and subcarinal regions. </think>

Lymphadenopathy is present in the right supraclavicular, mediastinal, right hilar, posterior mediastinal, paravertebral, para-aortic, and retrocrural regions, with bulky right supraclavicular and subcarinal adenopathy.

User: What is the most likely differential for the right lung mass?

NV-REASON-CT:

<think> The question asks for likely causes of the right lung mass. I review the CT pattern, including the right lower lobe mass, satellite nodules, multifocal ground-glass opacities, extensive adenopathy, and destructive paraspinal soft tissue. This combination is most concerning for malignancy with nodal and possible metastatic disease. Infection or inflammatory disease could contribute to the ground-glass opacities, but would be less likely to explain the mass and extensive adenopathy together. </think>

The leading consideration is primary lung malignancy with nodal and metastatic disease. Metastatic disease from another primary is another possibility. Infection or inflammatory disease may contribute to the multifocal ground-glass opacities, but the mass pattern and extensive adenopathy are more concerning for malignancy.

D.2. Chest CT reasoning output

User: full chest CT reasoning analysis

NV-REASON-CT:

<think> I will review the CT volume systematically, use the valid technique context, identify the key imaging findings, and then provide the final labels. </think>

We are going to review this IV contrast-enhanced CT of the chest in a systematic way. First, one important technical and support-device detail: there is a left-sided port catheter, and the catheter tip terminates in the superior vena cava, which is an expected central venous position.

I will begin with the lungs and airways. The central airways are patent, so there is no major tracheal or main bronchial occlusion. However, when we come down into the right lung, the right lower lobe is the dominant abnormality. There is a large mass-like process in the right lower lobe, and it is associated with partial collapse of the right lower lobe. This is a meaningful finding because a mass with postobstructive or associated volume loss is concerning for a malignant process.

Looking more carefully at the right lower lobe abnormality, it is not just simple atelectatic linear change. There is a moderate-sized mass component, and there are adjacent patchy opacities around it. Some of this could represent surrounding inflammatory change, postobstructive change, or adjacent small nodular disease. It is not easy to separate all of these components perfectly, but the overall morphology is clearly abnormal and concerning.

In the same abnormal right lower lobe region, there are a few small calcifications associated with the mass. Calcification can sometimes be seen in benign processes, but in this setting, because there is a large mass and extensive thoracic adenopathy, these calcifications do not make the finding reassuring. They are probably best considered part of the same pathologic process rather than a separate benign feature.

Now we move to the left lung. There are several patchy ground-glass opacities in the left lung. These are not dense mass-like consolidations. The appearance is more nonspecific and could be inflammatory or infectious in nature. I would favor an inflammatory-type process based on the patchy ground-glass morphology, although in the context of a suspected malignancy, it is important to acknowledge that imaging alone cannot be completely definitive.

At the apices, there is biapical scarring. This has a chronic-appearing pattern and is a common incidental-type finding. I

do not see this as the main clinically significant thoracic issue. There is no pneumothorax, and there is no pleural effusion, which is important because despite the large right lower lobe process, there is no associated pleural fluid collection or pleural air.

Next, we should evaluate the mediastinum and hilar regions. There is bulky mediastinal adenopathy. This includes enlarged lymph nodes in the right paratracheal region and subcarinal region. These are not just tiny incidental lymph nodes; they are bulky and clinically significant, and in the setting of the right lower lobe mass, they are concerning for nodal metastatic disease.

The right hilum is also abnormal, with bulky right hilar adenopathy. This is important because the dominant right lower lobe mass and right hilar nodal disease together create a pattern that is suspicious for thoracic malignancy with nodal involvement. The left lung has patchy ground-glass opacities, but the dominant nodal disease is on the right side.

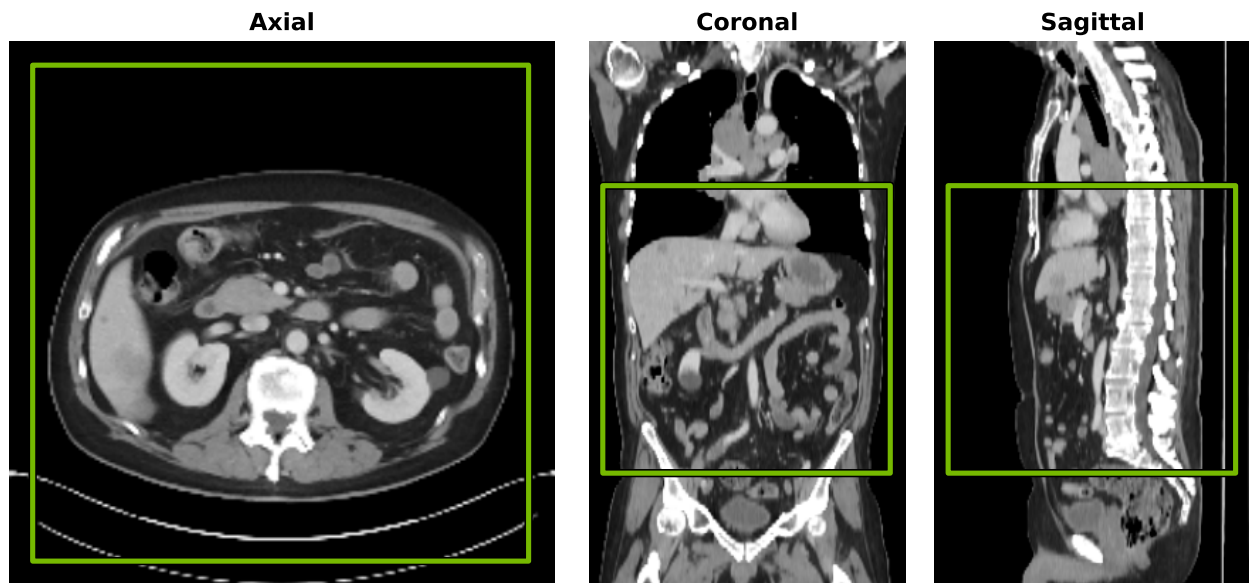
Now looking at the heart and pericardium, the heart size is within normal limits. There is no pericardial effusion. The thoracic aorta and main pulmonary artery are normal in caliber. There is a seven-vessel aortic arch, and the aortic root has a normal three-vessel branching pattern. I do not see an acute great-vessel abnormality.

The esophagus does not show a gross abnormality on this exam. The chest wall and breast soft tissues also do not show a discrete mass. So the major soft tissue abnormality is really centered in the right lower lobe lung and the mediastinal and right hilar lymph node stations rather than the superficial chest wall.

Finally, we need to check the bones carefully. There are multiple sclerotic lesions involving the thoracic spine, including lesions at the upper thoracic vertebral body region and lower thoracic vertebral body region. In this imaging context, these are concerning for osseous metastatic disease. It is also important to stress that there is no visible paravertebral soft tissue component, so there is no obvious extending extraosseous mass along those lesions.

Overall, the most significant thoracic findings are a large right lower lobe mass with partial right lower lobe collapse, bulky right hilar and mediastinal adenopathy, and sclerotic thoracic vertebral lesions concerning for metastatic involvement. There are also several patchy left lung ground-glass opacities, favored to be inflammatory or infectious in appearance. The overall thoracic pattern is suspicious for malignancy with nodal disease and osseous metastases.

Next, we show the abdominal crop example from the same full CT volume. The green box marks the square projection of the anatomy-aware $384 \times 384 \times 384$ -mm abdominal crop used as model input. The structured abdominal CT report and abdominal CT reasoning output follow.



D.3. Structured abdominal CT report

User: write a structured abdominal CT report

NV-REASON-CT:

<think> I will produce a structured abdominal CT report with a concise impression. </think>

TECHNIQUE: IV contrast CT.

FINDINGS:

Medical devices: None present.

ABDOMEN:

Liver and biliary system: Multiple hypoattenuating foci within the liver, including a small lesion within the left hepatic lobe and larger lesions within the right hepatic lobe, compatible with metastasis. No biliary ductal dilation. Gallbladder is unremarkable.

Spleen: Normal spleen.

Pancreas: Normal pancreas. No pancreatic ductal dilation.

Adrenal glands: Enhancing bilateral adrenal masses, including a moderate-sized right adrenal mass and a moderate-sized left adrenal mass.

Kidneys and urinary tract: Right lower pole renal cyst. Additional small hypodensities within the kidneys are too small to fully characterize. No urinary calculus or hydronephrosis.

Bowel and appendix: No evidence of bowel obstruction or inflammation.

Peritoneum and omentum: Peritoneal nodule in the left upper quadrant. No ascites or pneumoperitoneum.

Lymph nodes: Multiple prominent mesenteric lymph nodes, including a small mesenteric lymph node, reactive versus metastatic.

Vessels: Normal caliber abdominal aorta. The major vascular structures are patent. No abdominal vascular calcification or thrombosis.

Abdominal wall and soft tissues: No abdominal wall hernia or soft tissue mass.

Bones: Aggressive appearing lytic lesion within the right iliac bone involving the sacroiliac joint. Additional lytic lesion within the left iliac bone.

IMPRESSION:

1. Aggressive appearing right iliac lytic lesion involving the sacroiliac joint, highly suspicious for metastasis. Additional left iliac lytic lesion.
2. Enhancing bilateral adrenal masses compatible with metastasis.
3. Multiple hypoattenuating hepatic lesions compatible with metastasis.
4. Peritoneal nodule in the left upper quadrant.
5. Multiple prominent mesenteric lymph nodes, reactive versus metastatic.

User: Are the liver lesions solitary or multiple?

NV-REASON-CT:

<think> The question asks about the number and distribution of liver lesions. Multiple hypoattenuating liver lesions on contrast CT can suggest metastases, especially when distributed in both lobes. I inspect the CT image and see multiple non-cystic hypoattenuating foci. They involve both the right and left hepatic lobes, with a dominant right hepatic lesion. </think>

Multiple. The liver lesions are distributed in both the right and left hepatic lobes, with a dominant lesion in the right lobe.

User: What is the most likely differential for the liver lesions?

NV-REASON-CT:

<think> The question asks for the leading diagnostic possibilities for the liver lesions. I review the CT pattern, including multiple hypoattenuating liver lesions, enhancing adrenal masses, peritoneal nodule, and lytic iliac bone lesions. On CT, this combination most strongly suggests metastatic malignancy. Multifocal primary hepatic malignancy or inflammatory lesions would be less favored from the overall pattern alone. </think>

The leading consideration is metastatic malignancy, given the multiple liver lesions plus adrenal, peritoneal, and iliac bone lesions. Less likely possibilities, depending on clinical context, could include multifocal primary hepatic malignancy or inflammatory lesions, but the CT pattern most favors metastases.

User: which associated findings support it?

NV-REASON-CT:

<think> The question asks how the associated findings strengthen the suspected diagnosis. I review the CT beyond the liver lesions. The CT shows bilateral enhancing adrenal masses, a left upper quadrant peritoneal nodule, prominent mesenteric lymph nodes, and iliac lytic bone lesions. This multifocal pattern across organs and bones is highly suspicious for metastatic disease. The answer should describe the pattern without overclaiming tissue diagnosis. </think>

The associated findings that support suspected metastatic disease include bilateral enhancing adrenal masses, a left upper quadrant peritoneal nodule, prominent mesenteric lymph nodes, and lytic iliac bone lesions. CT alone cannot provide histology, but the pattern is concerning for metastases.

D.4. Abdominal CT reasoning output

User: full abdominal CT reasoning analysis

NV-REASON-CT:

<think> I will review the CT volume systematically, use the valid technique context, identify the key imaging findings, and then provide the final labels. </think>

We are reviewing an IV contrast-enhanced CT of the abdomen and pelvis. I do not see any medical device in the imaged abdomen. We will go through this in a systematic abdominal search pattern, starting with the solid organs and then moving to bowel, peritoneum, vessels, lymph nodes, abdominal wall, and bones.

First, looking at the liver, there are multiple focal lesions involving both hepatic lobes. The dominant abnormality is in the right hepatic lobe, and it has a central hypoattenuating component. That morphology is meaningful because it does not

look like a simple cyst; it has a masslike appearance with internal lower attenuation. In this context, the appearance is most suggestive of metastatic disease, with possible central necrotic change.

Now we check the biliary system and gallbladder. Despite the liver lesions, I do not see intrahepatic or extrahepatic biliary ductal dilatation. The gallbladder appears unremarkable, without imaging features to suggest an acute inflammatory gallbladder process.

The spleen appears unremarkable. Moving to the pancreas, there is a mass in the peripancreatic region. It is important to stress that there is no pancreatic ductal dilatation, but the soft tissue mass itself is still a clinically significant finding, especially with the other metastatic-appearing abnormalities in the abdomen.

Next, we evaluate the adrenal glands. Both adrenal glands contain masses. In isolation, adrenal lesions can sometimes be difficult to characterize, but when we consider the multiple liver lesions, peripancreatic mass, renal mass, and osseous lesions, these adrenal findings are very concerning for metastatic involvement.

Now we look at the kidneys and urinary tract. There is a mass involving the right kidney. I do not see hydronephrosis, and there is no urinary calculus. So even though there is a significant renal mass, there is no obstructive urinary pattern on this exam. The urinary tract otherwise does not show an acute obstructive process.

We then move to the bowel. I do not see evidence of bowel obstruction. That is important because despite the presence of peritoneal and mesenteric masses, the bowel loops are not dilated in a way that would suggest mechanical obstruction. There is no pneumoperitoneum, so there is no imaging evidence of free intraperitoneal air.

Now we focus on the peritoneum and mesentery. There are multiple peritoneal and mesenteric masses, including a dominant mass in the left upper abdomen. This pattern is very concerning for peritoneal metastatic disease. I do not see ascites, but the absence of ascites does not make these peritoneal masses less meaningful; the nodular masslike deposits themselves are the key finding.

Looking at the lymph node regions, there is a large right inguinal lymph node. In the setting of the other metastatic-appearing findings, this is also suspicious rather than incidental. The abdominal aorta is normal in caliber, and the abdominal vasculature appears patent.

The abdominal wall and superficial soft tissues do not show an abdominal wall hernia or soft tissue mass. Finally, we check the bones. There are multiple lytic and sclerotic osseous lesions, including involvement of the L2 vertebral body and a small sclerotic lesion in the right sacrum. The mixed lytic and sclerotic pattern is concerning for osseous metastatic disease. I do not see an acute fracture.

Overall, the most significant findings are multiple metastatic-appearing lesions involving the liver, a peripancreatic mass, bilateral adrenal masses, a right renal mass, multiple peritoneal and mesenteric masses, a large right inguinal lymph node, and mixed lytic and sclerotic bone lesions. The overall imaging pattern is most consistent with widespread abdominal and osseous metastatic disease.

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